

Aim

Conclusions

1. The two callers
2. Do the two callers agree
3. What the callers do on a donor with no cancer
4. Short read against long read
5. Per-donor detail
6. What was run
7. Data locations
8. Caveats

Benchmarking: calling cancer cells

Code ▾

2026-08-13

Runs: pilot 260423_VH01624_453_222HWMYX (HBDN206-MNpCT, HBDN392-AML-MDS, HBDN501-AML-KMT2A, MOLM13) and 260528_VH01624_464_222K7VKNX / LK2 (normal-01, HBDN376-TP53, HBDN498-TP53, APOP576-TP53).

Analysis finished: 2026-08-12 - the most recent of the 49 figures shown below.

Report generated: 2026-08-13 by 09.cancer_calling_report.Rmd , from the outputs of steps 01-08 in this folder.

Which cells in a patient sample are the leukaemia. There is no ground truth, so the question is approached by running two independent CNV callers on the same cells and asking where they agree, running one of them on a healthy donor to measure the false-positive floor, and running the same caller on short-read and long-read sequencing of the same cells to see how much the call depends on the data rather than the method.

For full report, download cancer_calling_report.html in the attachment and open in a browser.

Code used - attached and in the github repository: https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/8.Benchmarking-calling_cancer_cells (https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/8.Benchmarking-calling_cancer_cells)

Aim

Establish how malignant cells are identified, and how far the calls can be trusted:

- do two independent CNV callers agree on the same cells, and where do they not
- what does a caller do on a donor with no cancer - the false positive floor
- does the call depend on the sequencing platform
- which populations are hardest to call

Conclusions

- **CopyKAT and Numbat agree on the great majority of cells** - 96.35% and 94.96% on HBDN206-MNpCT and HBDN501-AML-KMT2A
- **but that headline agreement is inflated and should not be quoted on its own.** It is carried by HSC MPP, which is 4,002 of the 4,409 shared cells in HBDN206-MNpCT and agrees 99.8%. Agreement in the lymphoid compartment is far worse - Naive T 25%, B 51%, CD4 Memory T 63% - and those are exactly the populations handed to CopyKAT as its confident-normal anchor. CopyKAT calls them diploid partly by construction, so the two callers are not independent there
- **where they disagree, it is one-directional: Numbat calls malignant, CopyKAT calls normal** (152 against 9 cells in HBDN206-MNpCT, 73 against 2 in HBDN501-AML-KMT2A). Numbat is the more liberal caller on this data
- **Numbat is specific.** On the healthy donor its CNV evidence sits at a median `cnv_z_abs_mean` of 15.3 against 123-315 in the three AML donors, and only **1.0%** of normal-01 cells exceed the 99th percentile of the normal distribution against **97-100%** of AML cells. Calling cancer where there is none is not the failure mode to worry about
- **the platform changes the answer more than the caller does.** Same cells, same Numbat, short read against long read: compartment concordance is 94.2%, 89.5%, 63.6%, but per-cell pCNV correlation is only 0.19, 0.41, 0.12. So the two platforms sort cells into the same bin reasonably often while disagreeing almost entirely about how much CNV evidence each cell carries
- **that disagreement is also one-directional** - short read calls tumour where long read calls normal (339, 566 and 365 cells) and almost never the reverse (0, 99, 2). Long read is the more conservative platform, which is expected given its much lower depth per cell
- **Numbat did not complete for two of the four pilot donors.** HBDN392-AML-MDS and MOLM13 are 100% `not_called` . Any pilot statement about Numbat rests on two donors
- **MOLM13 is never actually called by either method.** It is an AML cell line with no normal cells to anchor a baseline, so it is assigned aneuploid directly. It is flagged in its own column so it can be excluded from concordance statistics, and it should be

1. The two callers

CopyKAT and Numbat both infer copy number from single-cell RNA, but they use different evidence and need different inputs.

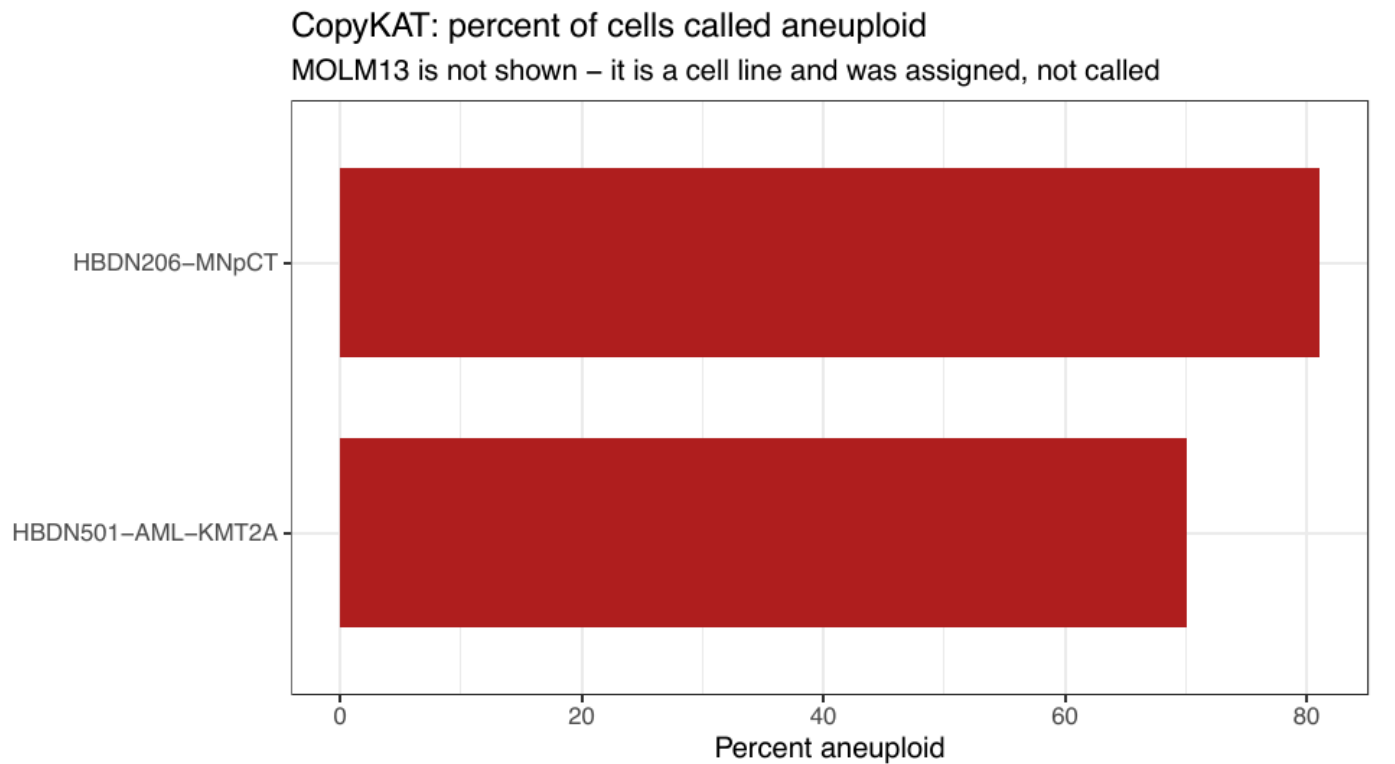
CopyKAT works from expression alone. It needs a set of cells it can trust to be diploid, and it is given the lymphoid populations - Naive T, CD4 Memory T, CD8 Memory T, NK and B - on the grounds that they are not part of the myeloid compartment the leukaemia arises from. It is run **per donor**, not per GEM well: the well is a pool of four donors and CopyKAT estimates its baseline from the cells it is handed, so pooling would let one donor's aneuploidy define another donor's normal.

Numbat adds allele evidence. It needs the Cell Ranger BAM, a phased SNP panel and a population reference panel, so it runs in two cluster stages inside the Numbat container - pileup and phasing first (step 02), then the CNV and clone inference (step 03).

CopyKAT calls per donor, LK1 pilot

donor	copykat_call	n_cells	total	percent
HBDN206-MNpCT	aneuploid	4619	5697	81.08
HBDN206-MNpCT	diploid	774	5697	13.59
HBDN206-MNpCT	not.defined	304	5697	5.34
HBDN501-AML-KMT2A	aneuploid	1965	2803	70.10
HBDN501-AML-KMT2A	diploid	498	2803	17.77
HBDN501-AML-KMT2A	not.defined	340	2803	12.13

01_copykat_aneuploid_by_donor.pdf



Numbat compartment per donor

run_label	sample	numbat_compartment	n_cells	total	percent
LK1	HBDN206-MNpCT	normal	150	5981	2.51
LK1	HBDN206-MNpCT	not_called	1428	5981	23.88
LK1	HBDN206-MNpCT	tumor	4403	5981	73.62
LK1	HBDN392-AML-MDS	not_called	6144	6144	100.00
LK1	HBDN501-AML-KMT2A	normal	11	3154	0.35
LK1	HBDN501-AML-KMT2A	not_called	1538	3154	48.76

run_label	sample	numbat_compartment	n_cells	total	percent
LK1	HBDN501-AML-KMT2A	tumor	1605	3154	50.89
LK1	MOLM13	not_called	6262	6262	100.00

Two donors come back entirely not_called . HBDN392-AML-MDS and MOLM13 did not produce usable Numbat output, so every Numbat statement in this report is based on HBDN206-MNpCT and HBDN501-AML-KMT2A only.

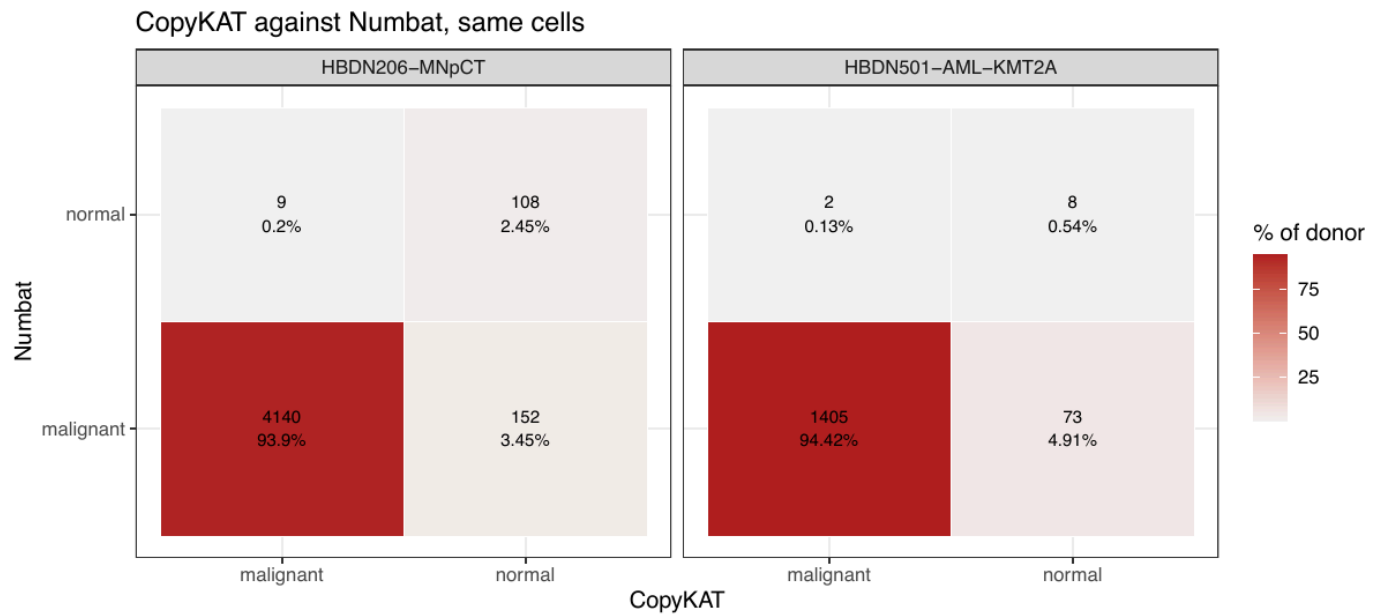
2. Do the two callers agree

Cell-level agreement, on cells with a call from both

donor	n_cells	percent_agree	n_numbat_only	n_copykat_only
HBDN206-MNpCT	4409	96.35	152	9
HBDN501-AML-KMT2A	1488	94.96	73	2

02_copykat_vs_numbat_confusion.pdf

The off-diagonal is asymmetric. Numbat calling malignant where CopyKAT calls normal is much more common than the reverse.



The headline number is high, and on its own it is misleading. Breaking agreement down by cell type shows where it actually comes from:

Agreement by broad cell type (populations with at least 20 cells)

predicted_CellType_Broad	n_cells	percent_agree
Naive T	20	25.0
B	41	51.2
Pro-Monocyte	57	54.4
CD4 Memory T	43	62.8
Monocyte	490	75.9
CD8 Memory T	69	79.7
Late GMP	230	96.5
HSC MPP	4002	99.8
Cycling Progenitor	26	100.0
Early GMP	281	100.0
EoBasoMast Precursor	87	100.0
LMPP	255	100.0

predicted_CellType_Broad	n_cells	percent_agree
MEP	226	100.0

HSC MPP is 4,002 of the 4,409 shared cells and agrees 99.8%, which is what drags the overall figure to 96%. The lymphoid populations - the ones the two callers actually have room to disagree about - sit between 25% and 80%.

This is worth being blunt about: **the lymphoid cells are CopyKAT's normal reference**. CopyKAT was told they are diploid, so its calling them diploid is not independent evidence. Where Numbat calls some of those same cells malignant, the disagreement cannot be settled by this comparison - it is a question about whether the normal anchor was correctly chosen, and that is what step 05's clone inspection is for.

3. What the callers do on a donor with no cancer

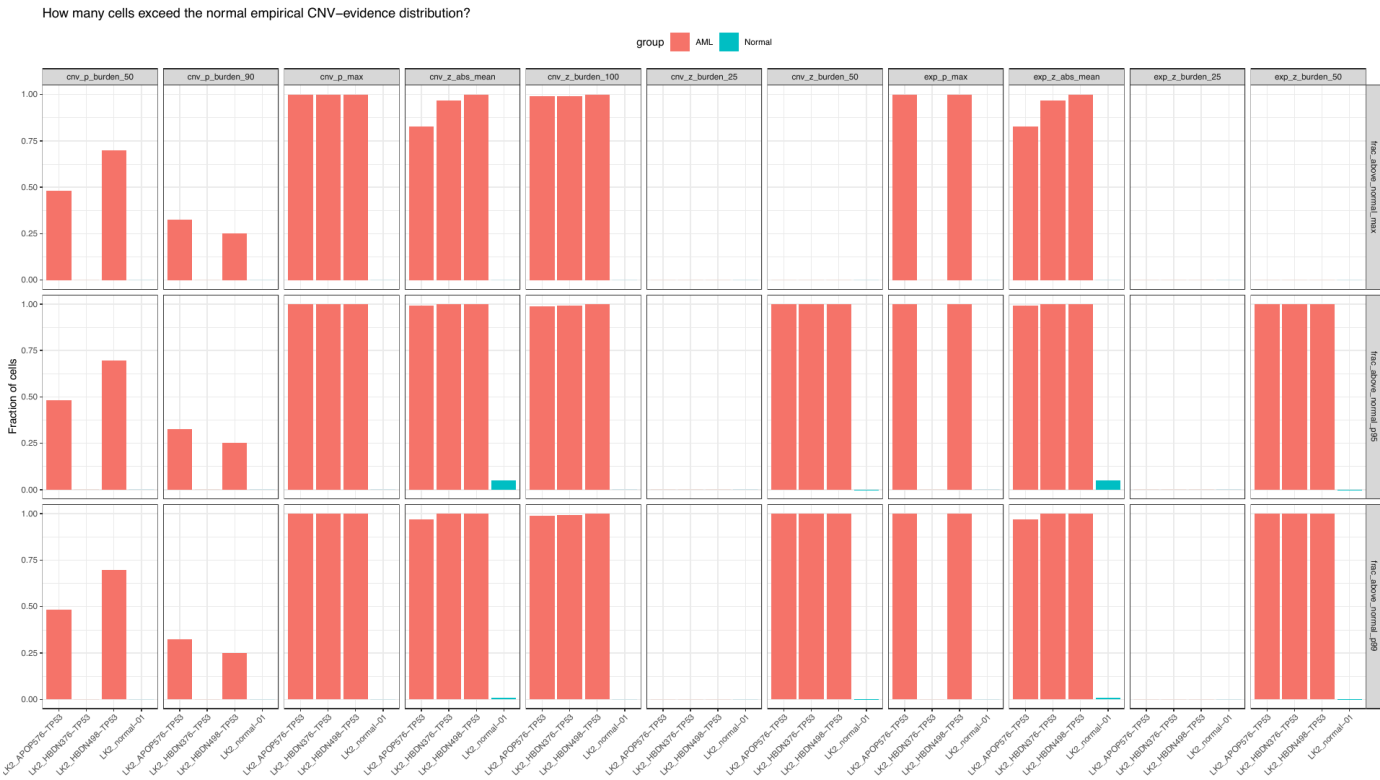
normal-01 in LK2 is a healthy donor. Anything called tumour there is a false positive, so it gives the floor that the AML numbers should be read against. Step 06 compares it against the three AML donors on the same per-cell CNV metrics.

Tumour-like fraction against the normal donor's own distribution

sample	group	metric	median	frac_above_normal_p99	frac_above_normal_max
LK2_APOP576-TP53	AML	cnv_z_abs_mean	133.514	0.971	0.827
LK2_HBDN376-TP53	AML	cnv_z_abs_mean	123.480	0.999	0.967
LK2_HBDN498-TP53	AML	cnv_z_abs_mean	315.191	1.000	1.000
LK2_normal-01	Normal	cnv_z_abs_mean	15.324	0.010	0.000
LK2_APOP576-TP53	AML	cnv_z_burden_100	0.538	0.990	0.990
LK2_HBDN376-TP53	AML	cnv_z_burden_100	0.500	0.991	0.991
LK2_HBDN498-TP53	AML	cnv_z_burden_100	0.810	1.000	1.000
LK2_normal-01	Normal	cnv_z_burden_100	0.000	0.000	0.000

02_fraction_above_normal_cnv_cutoffs.pdf

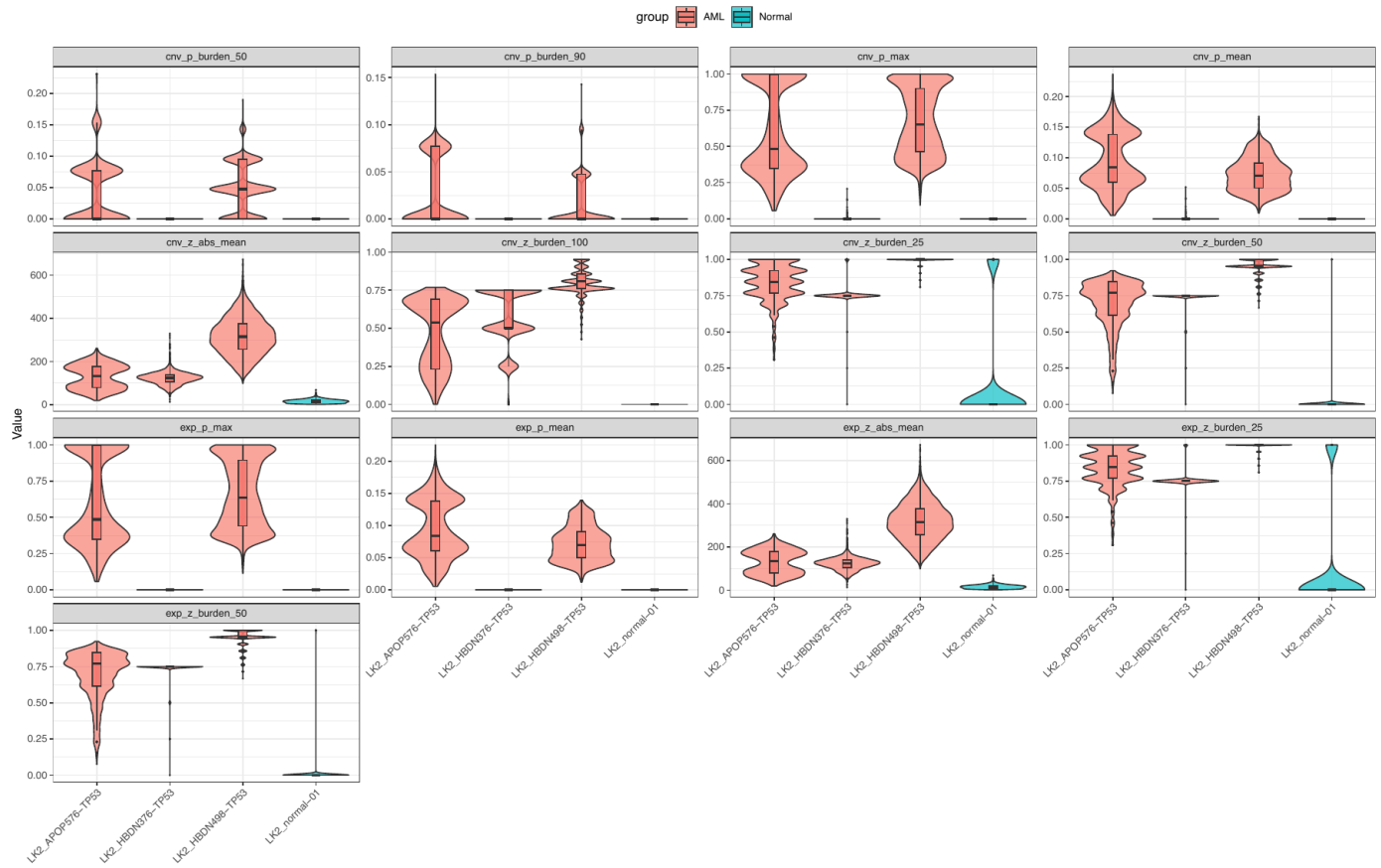
Fraction of each donor's cells exceeding the normal donor's cutoffs. The three AML donors sit at the top, normal-01 at the floor.



01_interpretable_per_cell_cnv_metrics.pdf

The per-cell metrics themselves, normal against AML.

Interpretable Numbat CNV evidence per cell
 Use CNV burden / altered-segment burden, not clone posterior, to judge whether normal cells are truly tumour-like

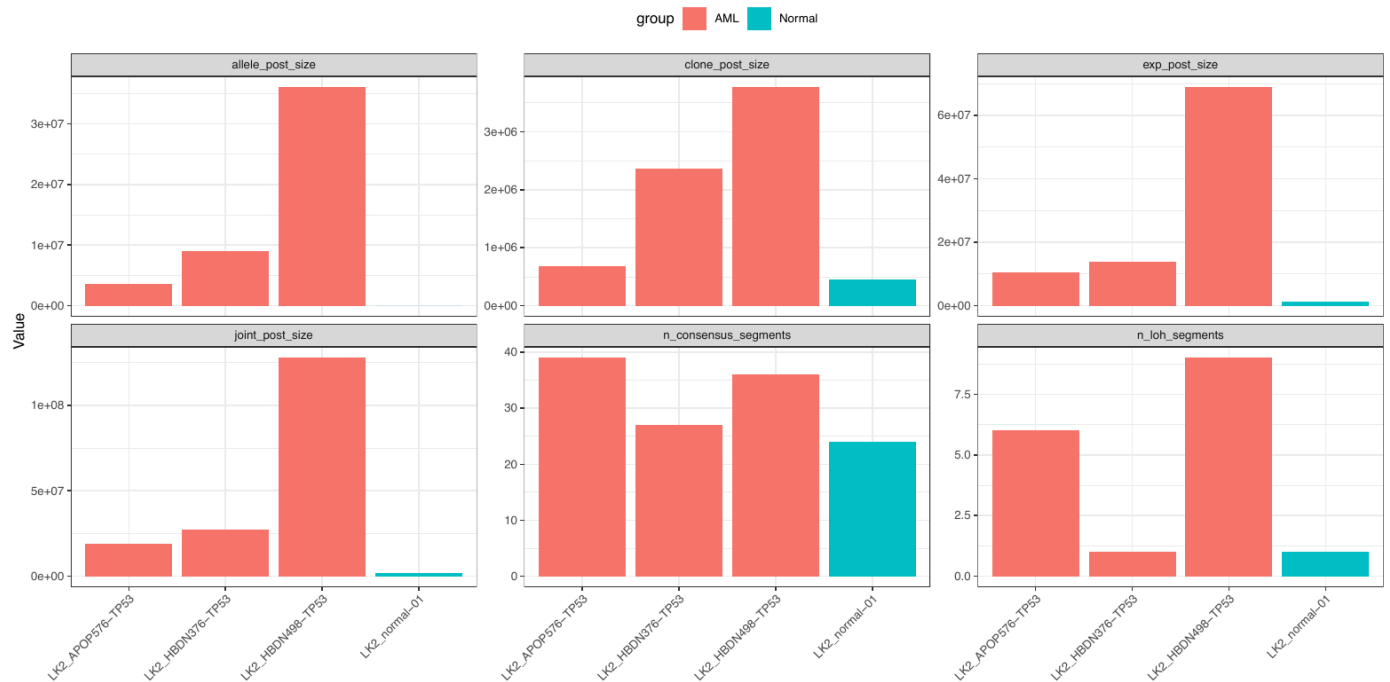


03_sample_level_evidence.pdf

Sample-level CNV evidence.

Sample-level Numbat evidence

Normal may have consensus segments, but should have much weaker allele/joint evidence



The separation is clean. Median **cnv_z_abs_mean** is 15.3 in normal-01 against 123, 133 and 315 in the AML donors, and 1.0% of normal cells exceed the normal 99th percentile against 97-100% of AML cells. Numbat is not manufacturing cancer out of a healthy marrow.

4. Short read against long read

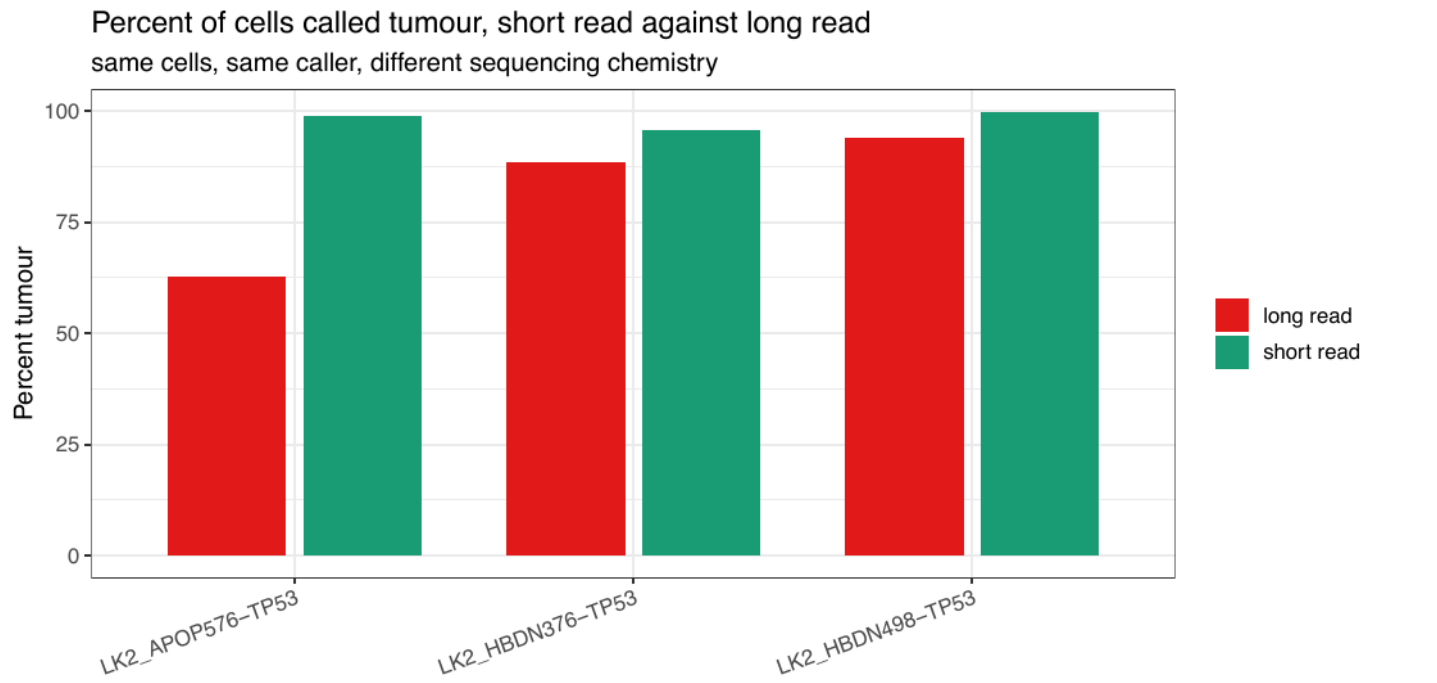
The same LK2 cells were sequenced on both platforms and Numbat was run on each. Same donors, same barcodes, same caller - so any disagreement is the data, not the method.

Short read against long read, per donor

donor	n_cells	sr_percent_tumor	lr_percent_tumor	compartment_concordance	pcnv_correlation	sr_tumor_lr_normal	sr_normal_lr_tumor
LK2_HBDN498-TP53	5790	99.81	93.96	94.15	0.189	339	0
LK2_HBDN376-TP53	6343	95.84	88.48	89.52	0.413	566	99
LK2_APOP576-TP53	1008	98.91	62.90	63.59	0.122	365	2

03_shortread_vs_longread_tumor_fraction.pdf

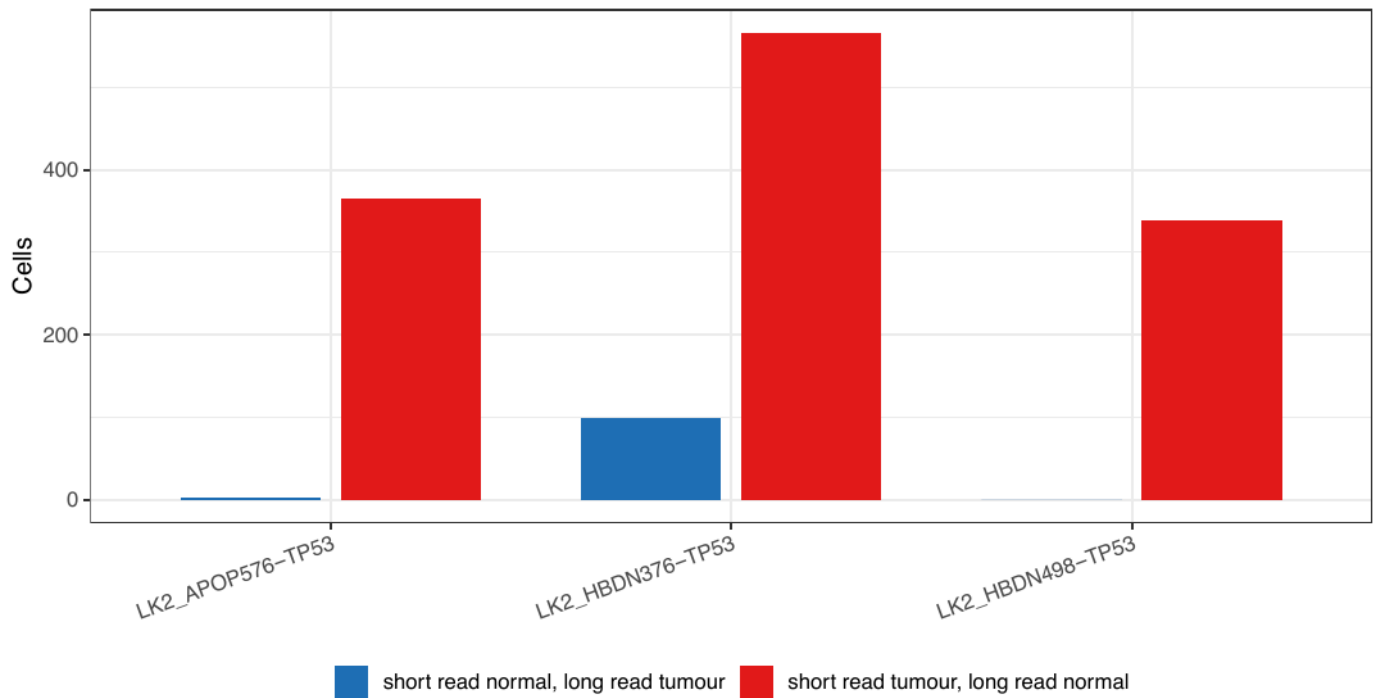
Long read calls fewer cells tumour in every donor.



04_shortread_vs_longread_direction.pdf

The disagreement is almost entirely one-directional. Cells that short read calls tumour and long read calls normal outnumber the reverse by orders of magnitude in two of the three donors.

Which way the two platforms disagree

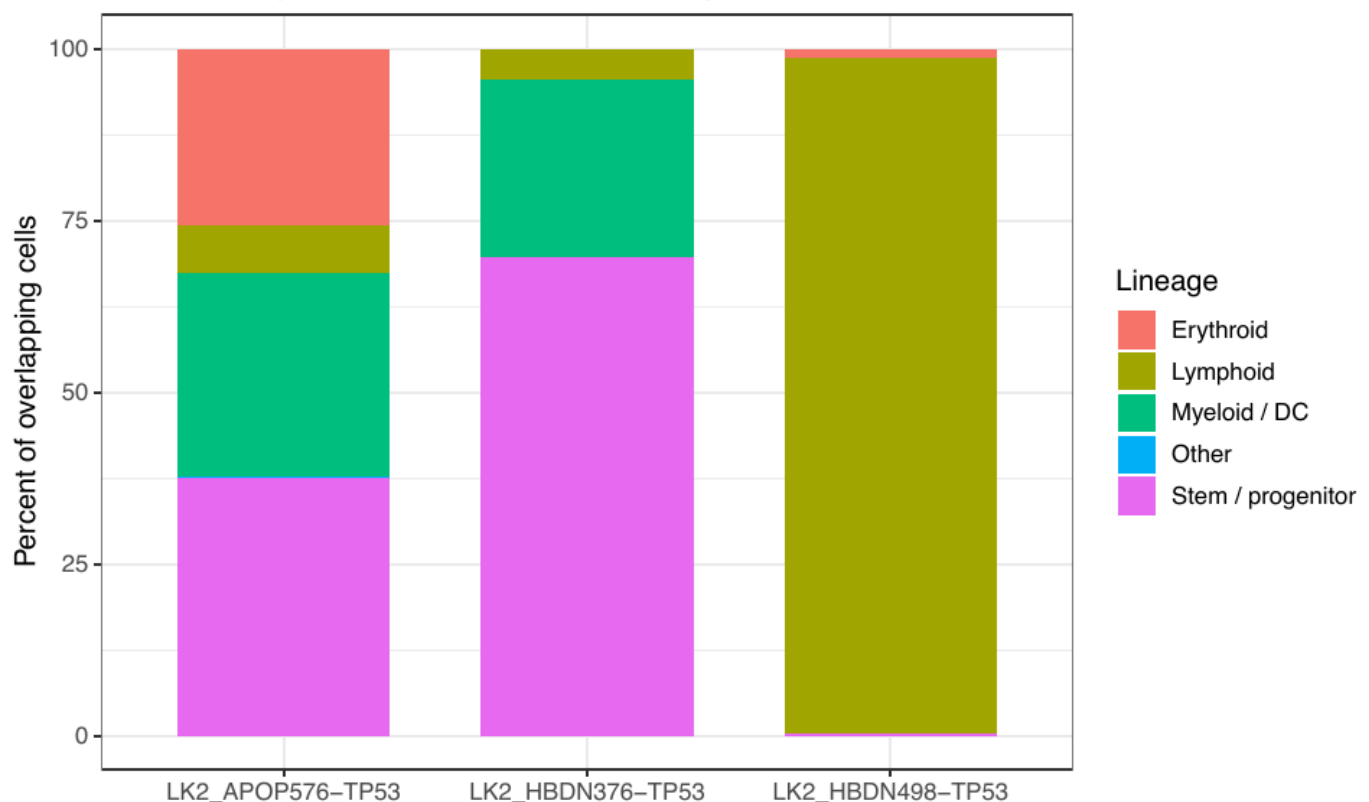


Two numbers should be read together here. Compartment concordance looks respectable at 89-94% for two of the donors, but the per-cell pCNV correlation is 0.12-0.41. The platforms put cells in the same bin without agreeing on the evidence behind it, which means the compartment call is robust in a way the per-cell CNV probability is not. Anything downstream that uses pCNV as a continuous quantity should not be trusted across platforms.

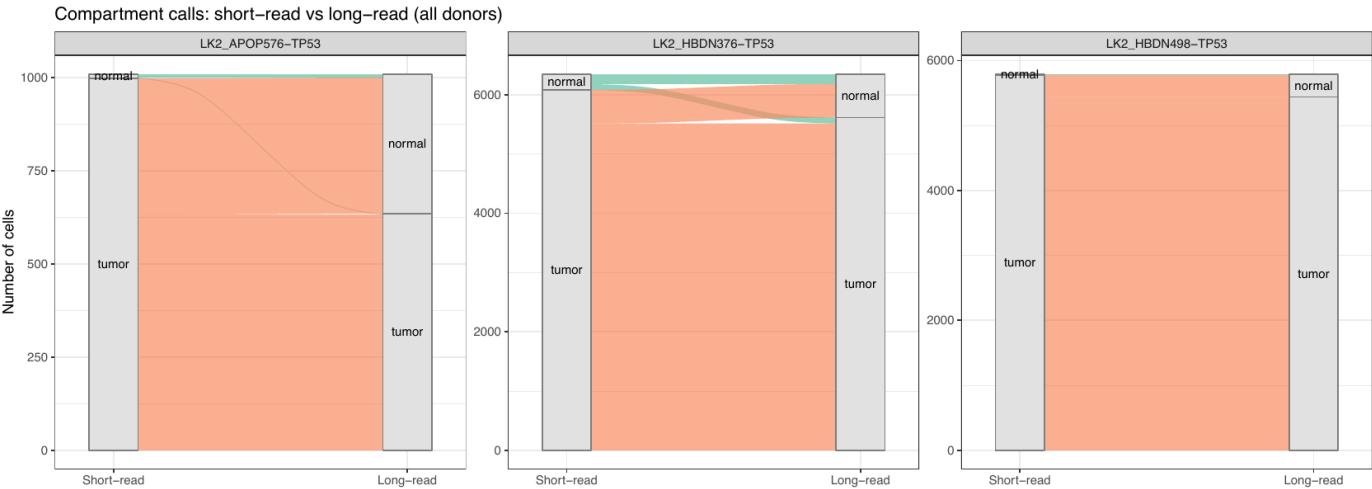
APOP576-TP53 is the worst on both counts - 63.6% concordance and 0.12 correlation. It is also the sample that loses 30% of its cells to BoneMarrowMap mapping-error QC in the annotation benchmark. The same sample being the outlier in two unrelated analyses points at the sample rather than at either method.

[summary_broad_lineage_composition_overlap.pdf](#)

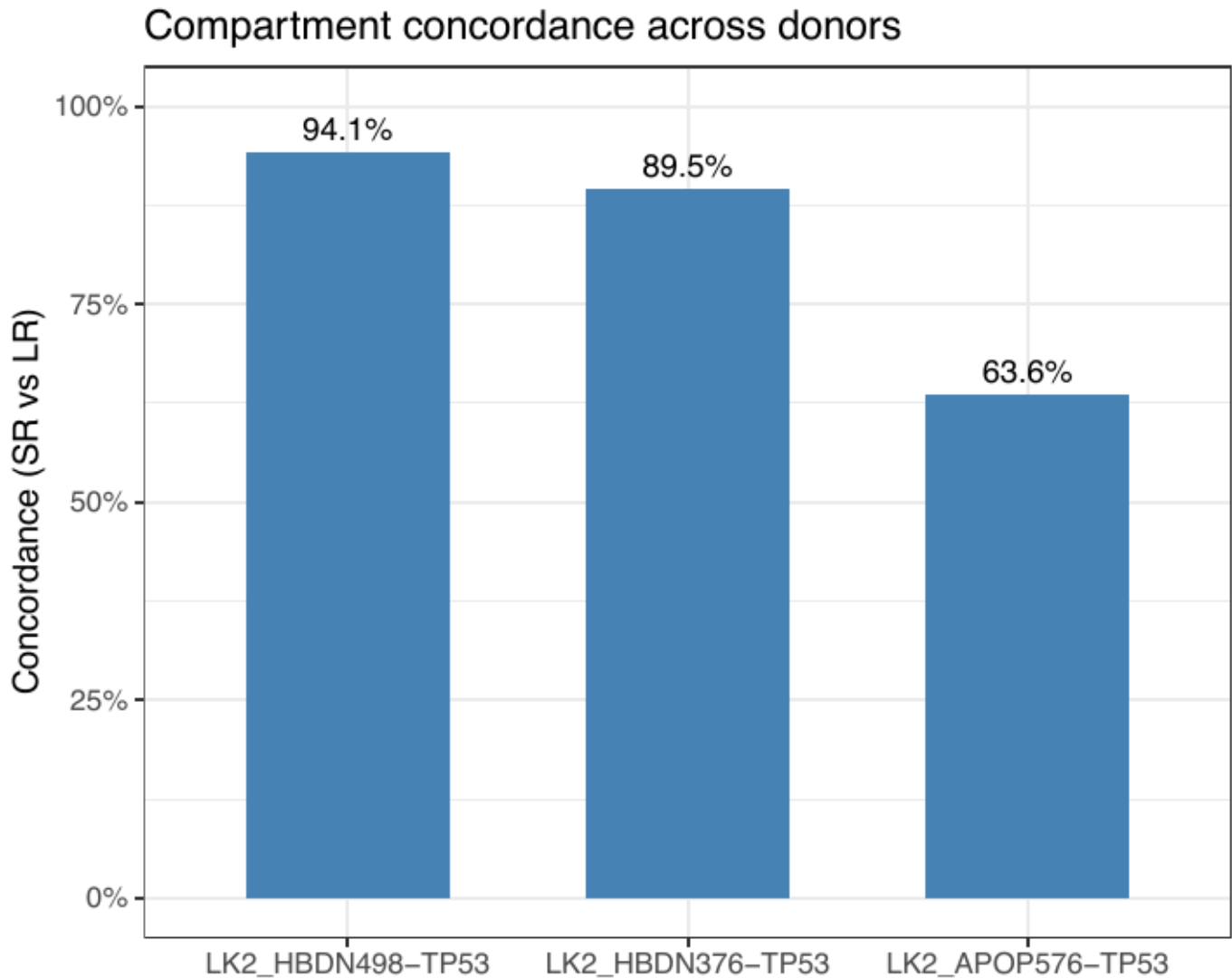
Broad lineage composition of overlapping cells



summary_compartment_alluvial_all.pdf

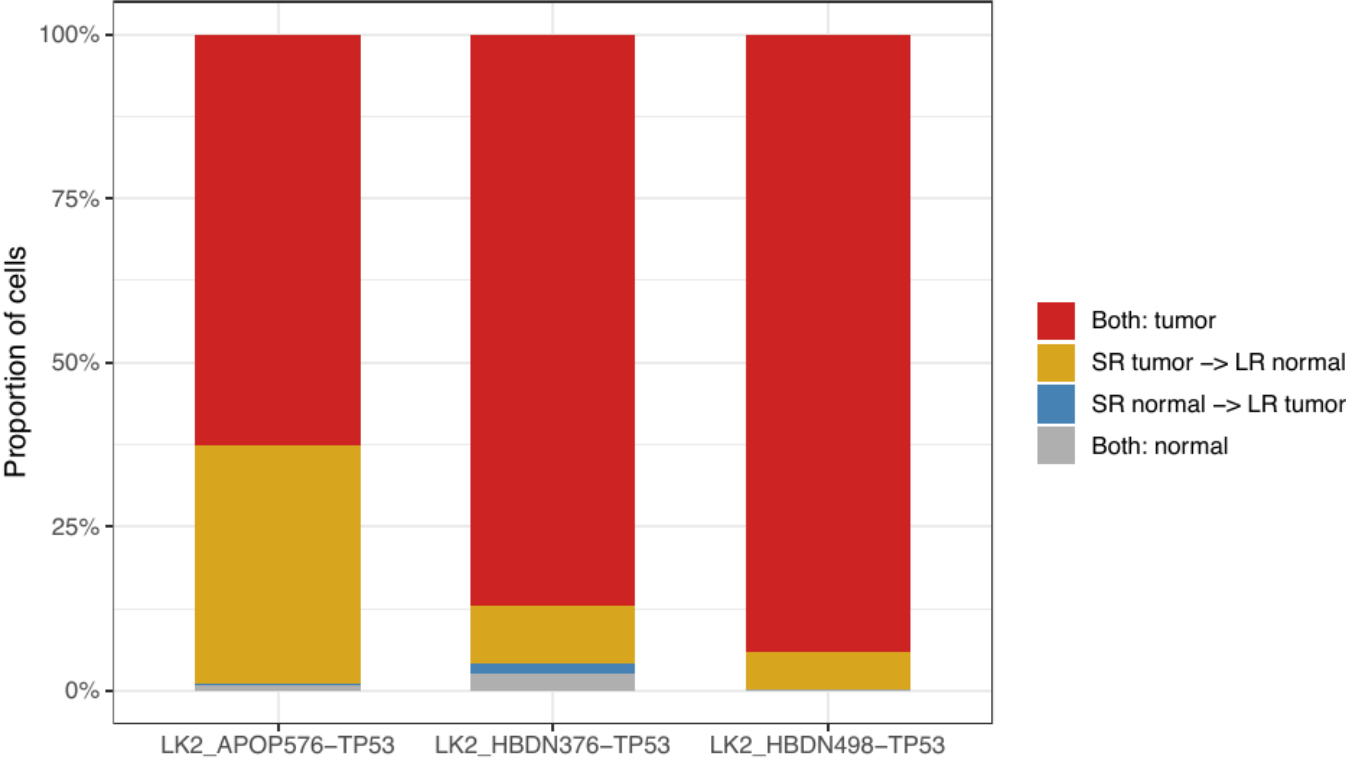


summary_compartment_concordance.pdf

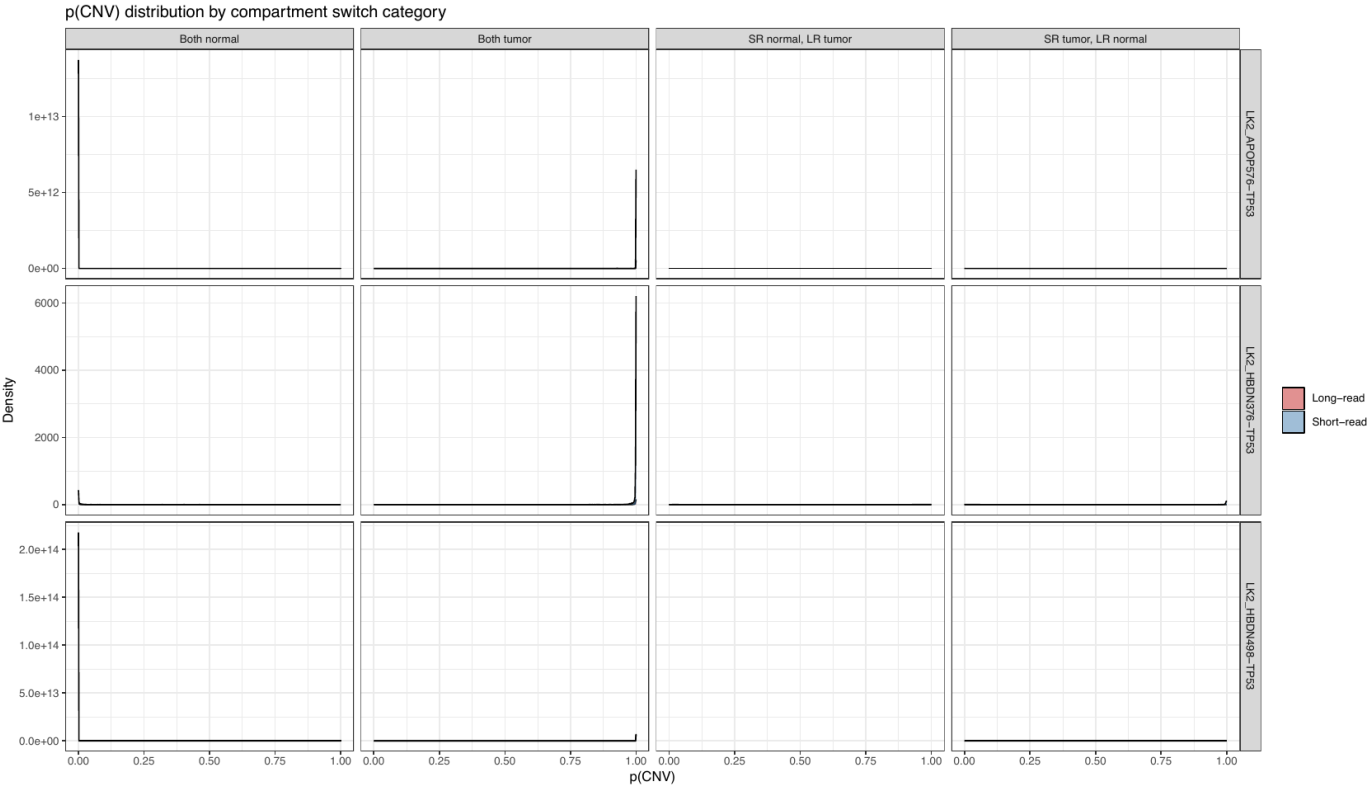


summary_compartment_switches.pdf

Compartment classification breakdown across donors

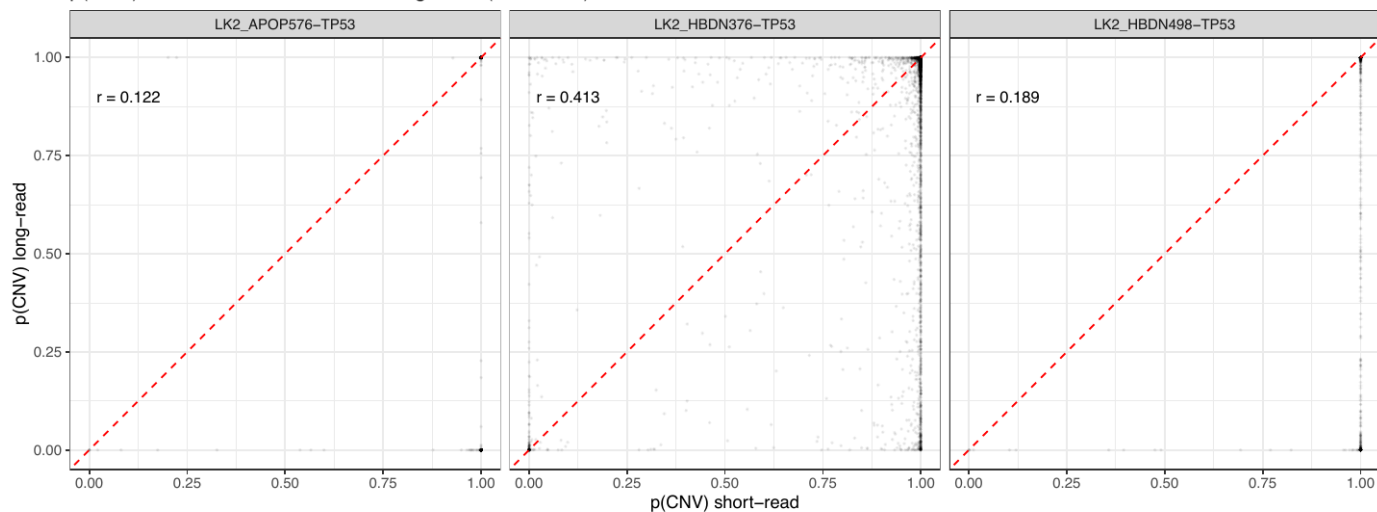


summary_pCNV_density_by_switch.pdf



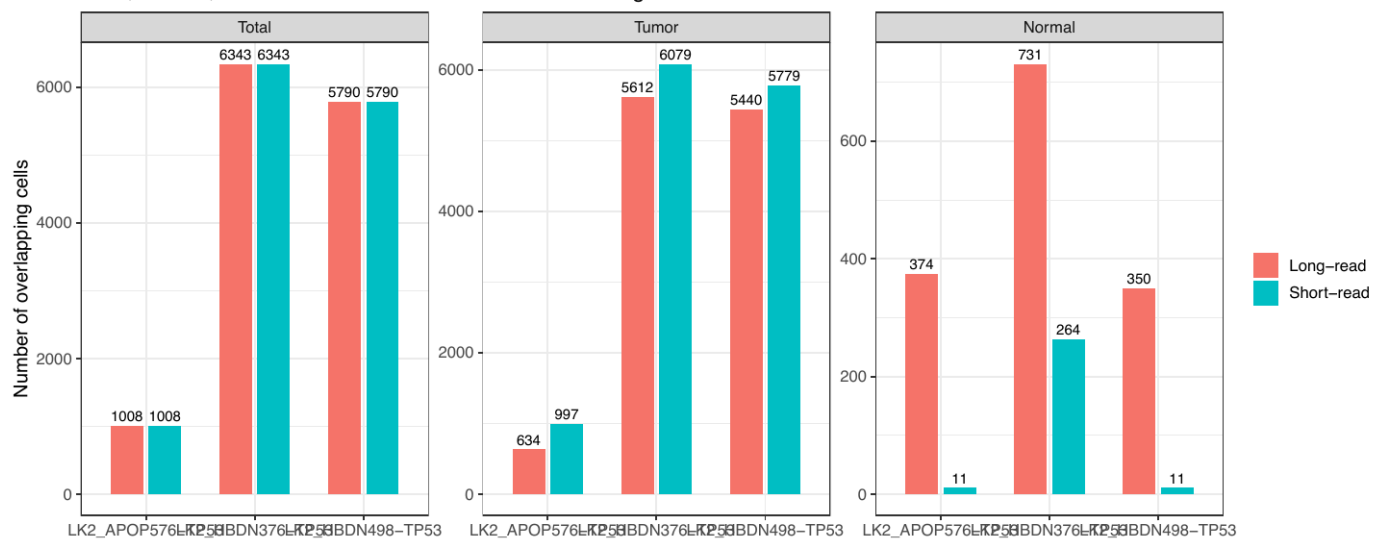
summary_pCNV_scatter_all.pdf

p(CNV) correlation: short-read vs long-read (all donors)



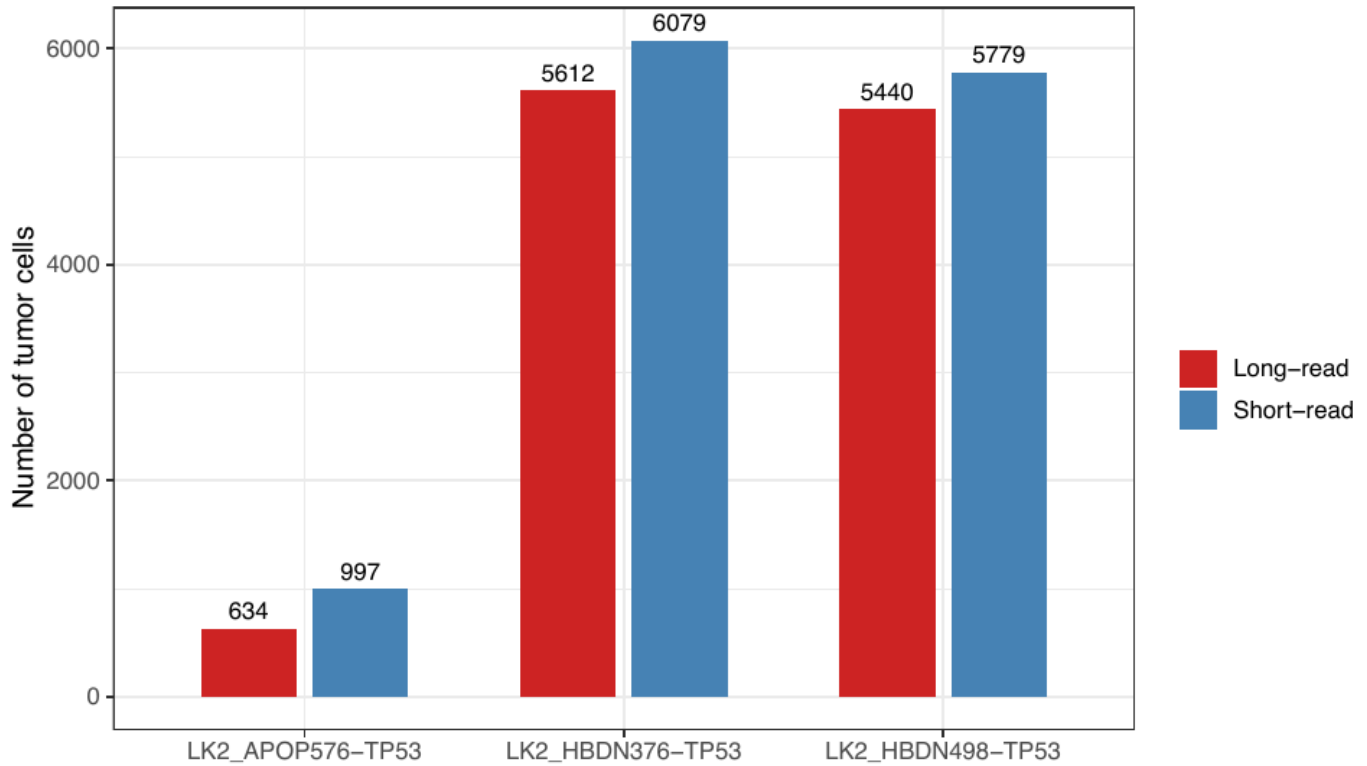
summary_total_tumor_normal_counts.pdf

Total, tumour, and normal cell counts: short-read vs long-read



summary_tumor_counts.pdf

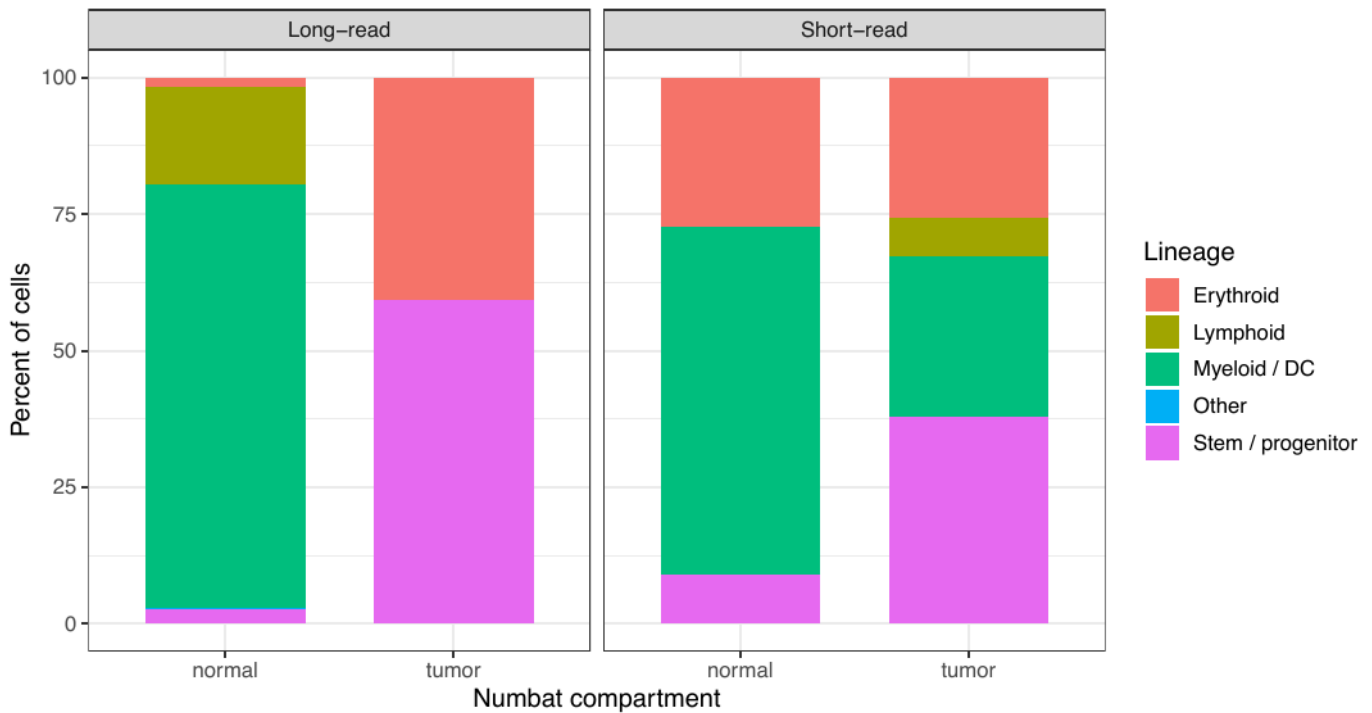
Tumor cell counts: short-read vs long-read



5. Per-donor detail

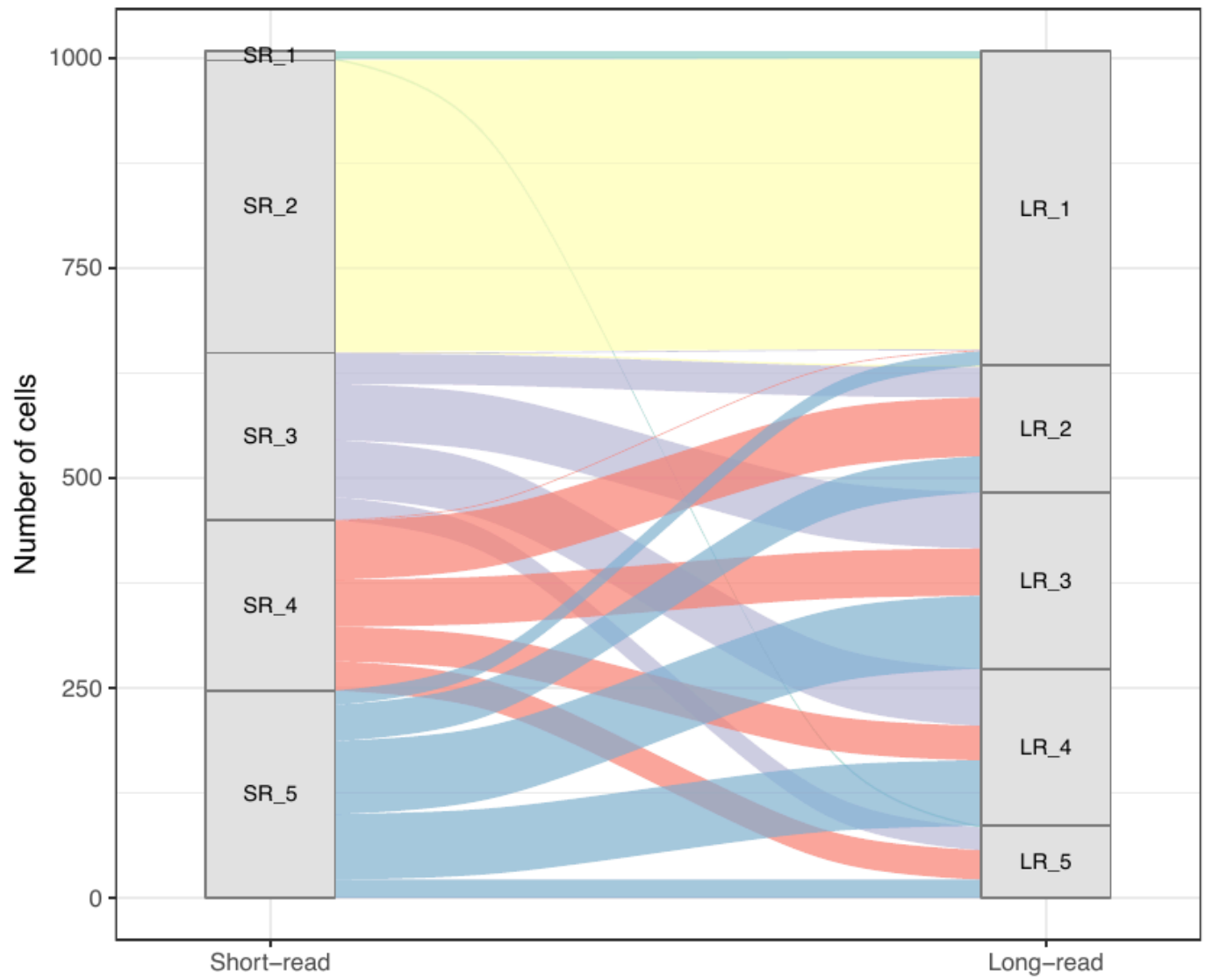
LK2_APOP576-TP53_broad_lineage_composition_by_compartment.pdf

LK2_APOP576-TP53 – Broad lineage composition by Numbat compartment



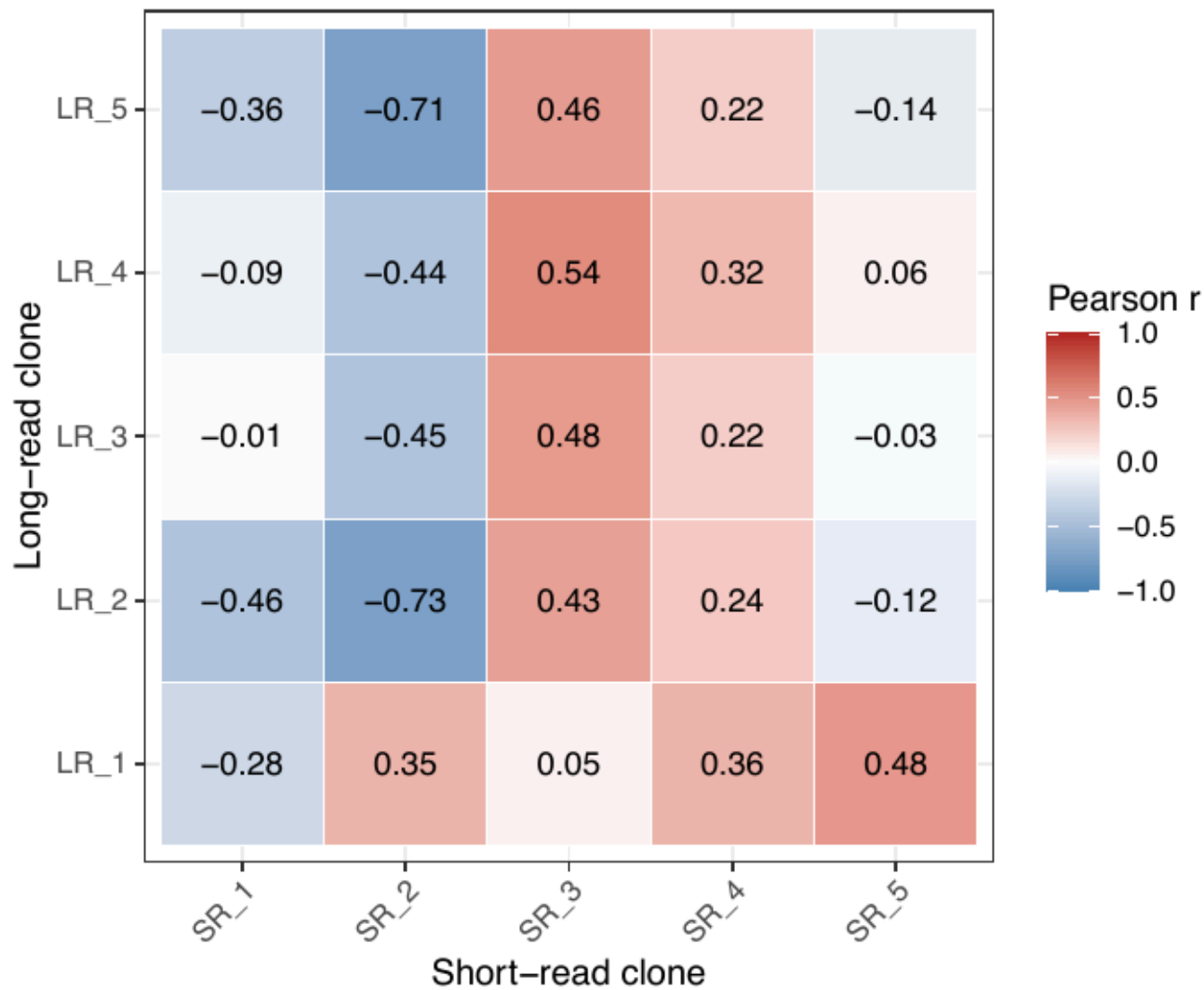
LK2_APOP576-TP53_clone_alluvial.pdf

LK2_APOP576-TP53 – Clone: short-read vs long-read

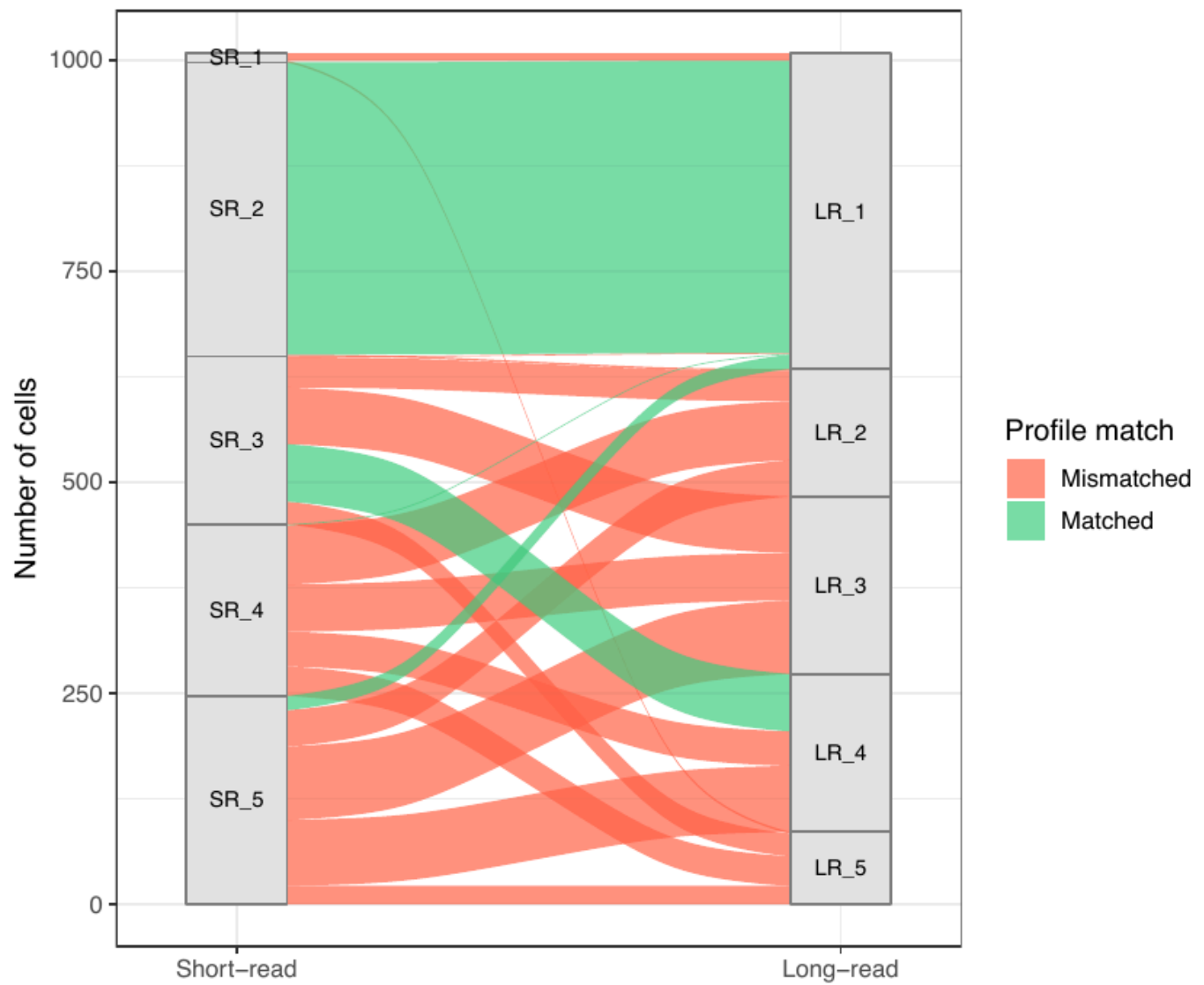


LK2_APOP576-TP53_clone_correlation_heatmap.pdf

LK2_APOP576-TP53 – Clone CNV profile correlation

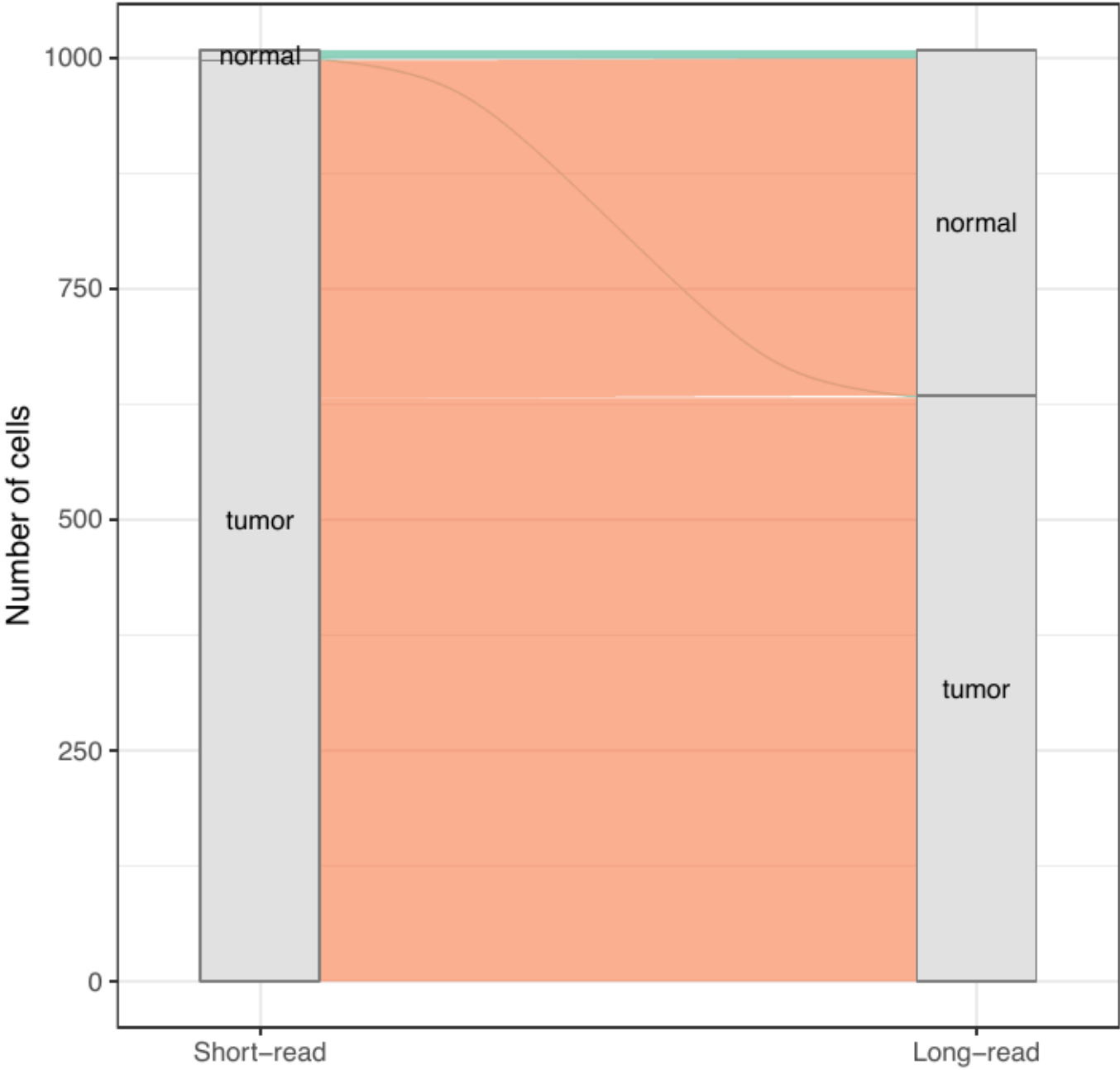


LK2_APOP576-TP53 – Clone matching (by CNV profile)

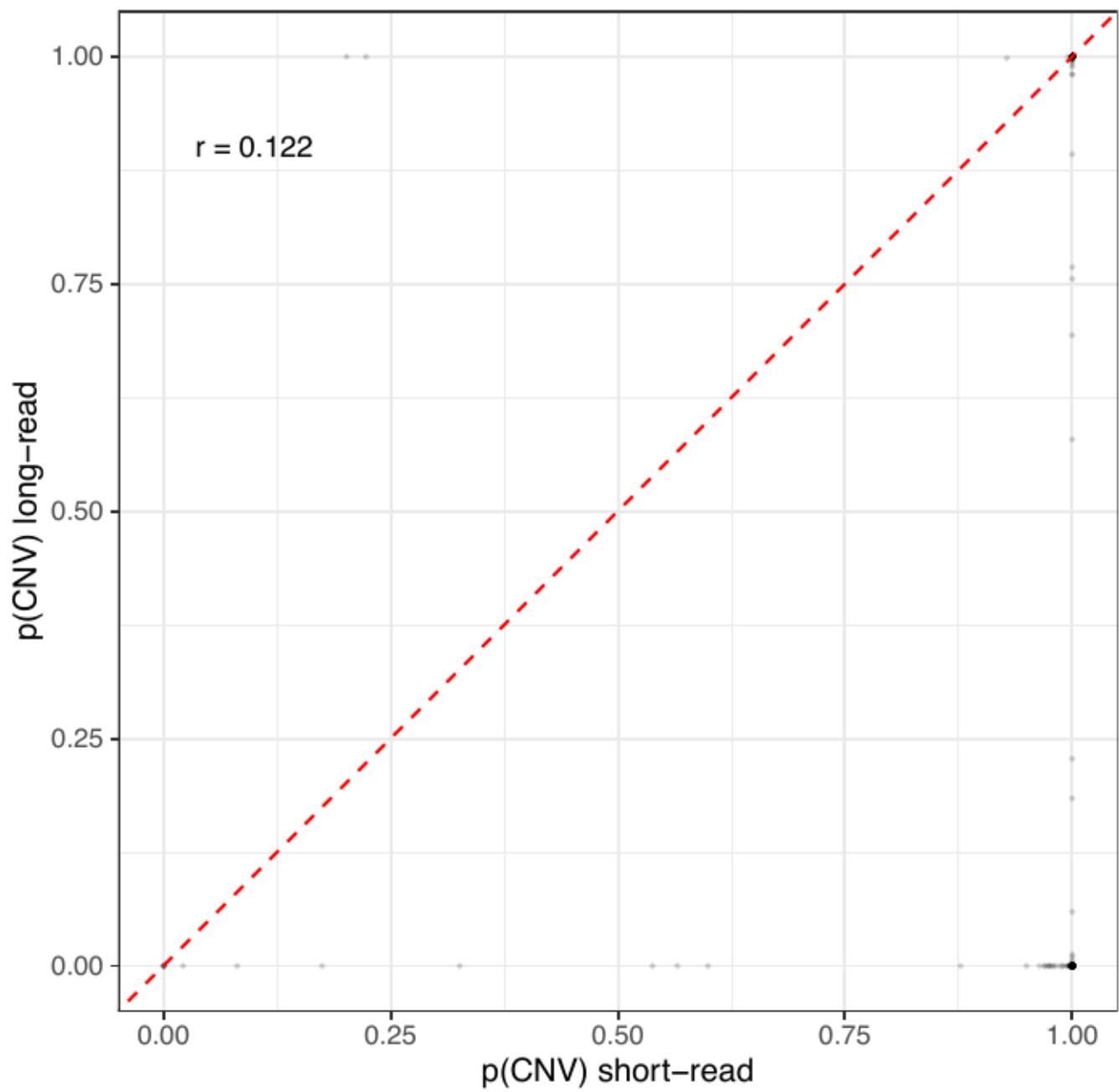


LK2_APOP576-TP53_compartment_alluvial.pdf

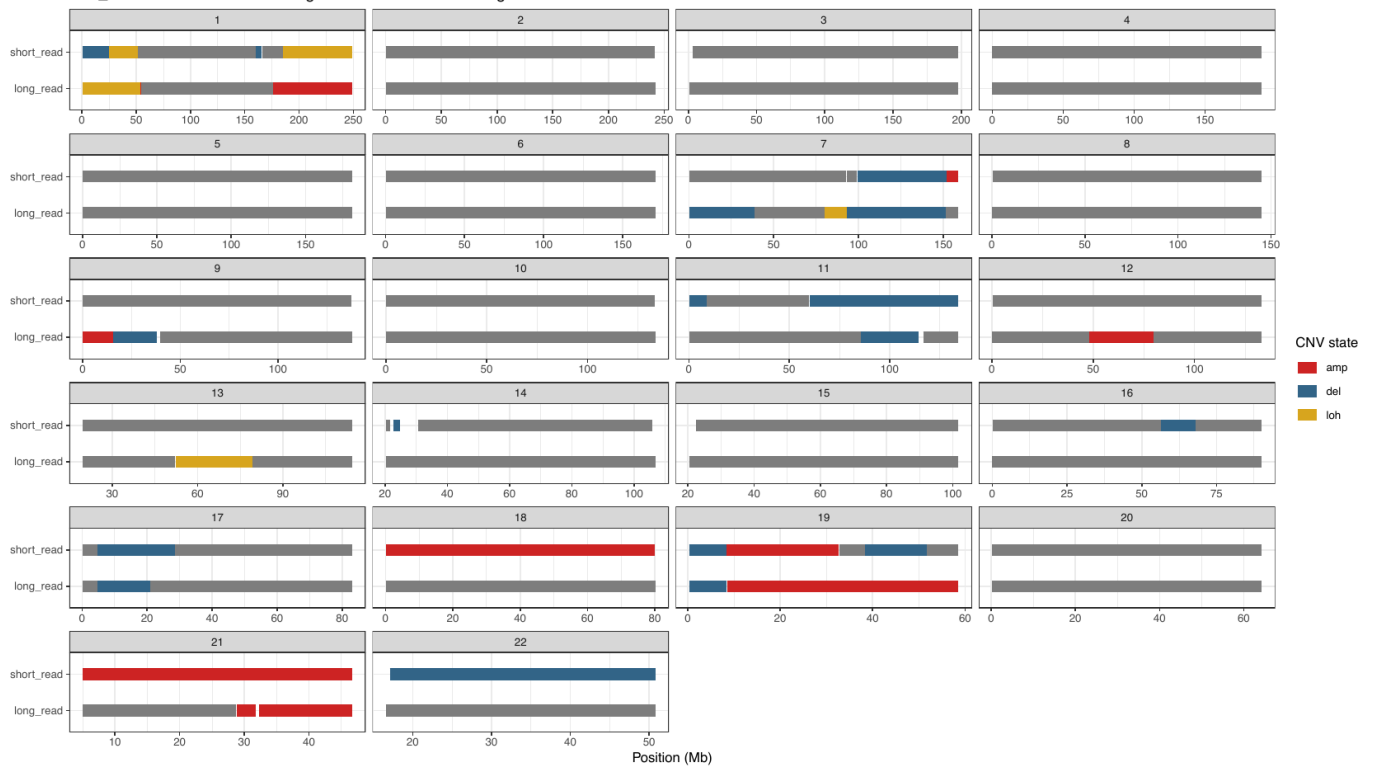
LK2_APOP576-TP53 – Compartment: short-read vs long-read



LK2_APOP576-TP53 – p(CNV) correlation

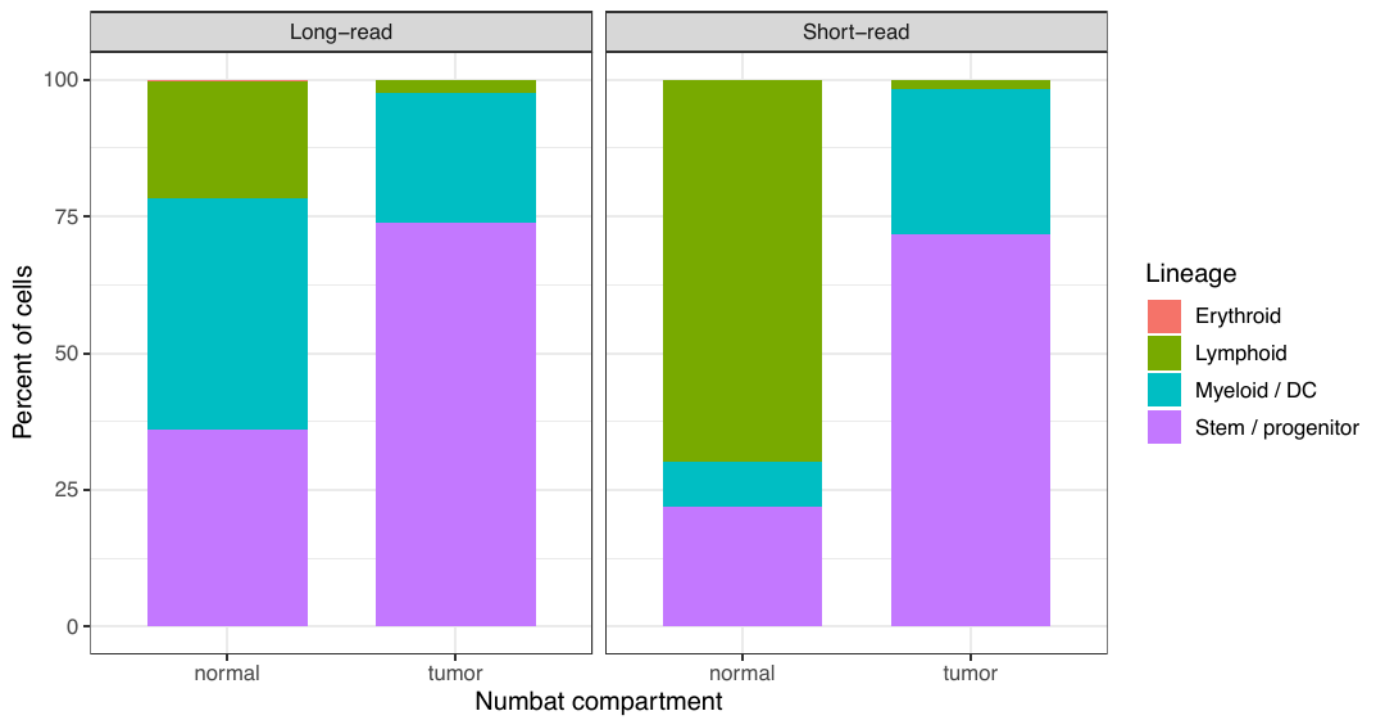


LK2_APOP576-TP53 – CNV segments: short-read vs long-read



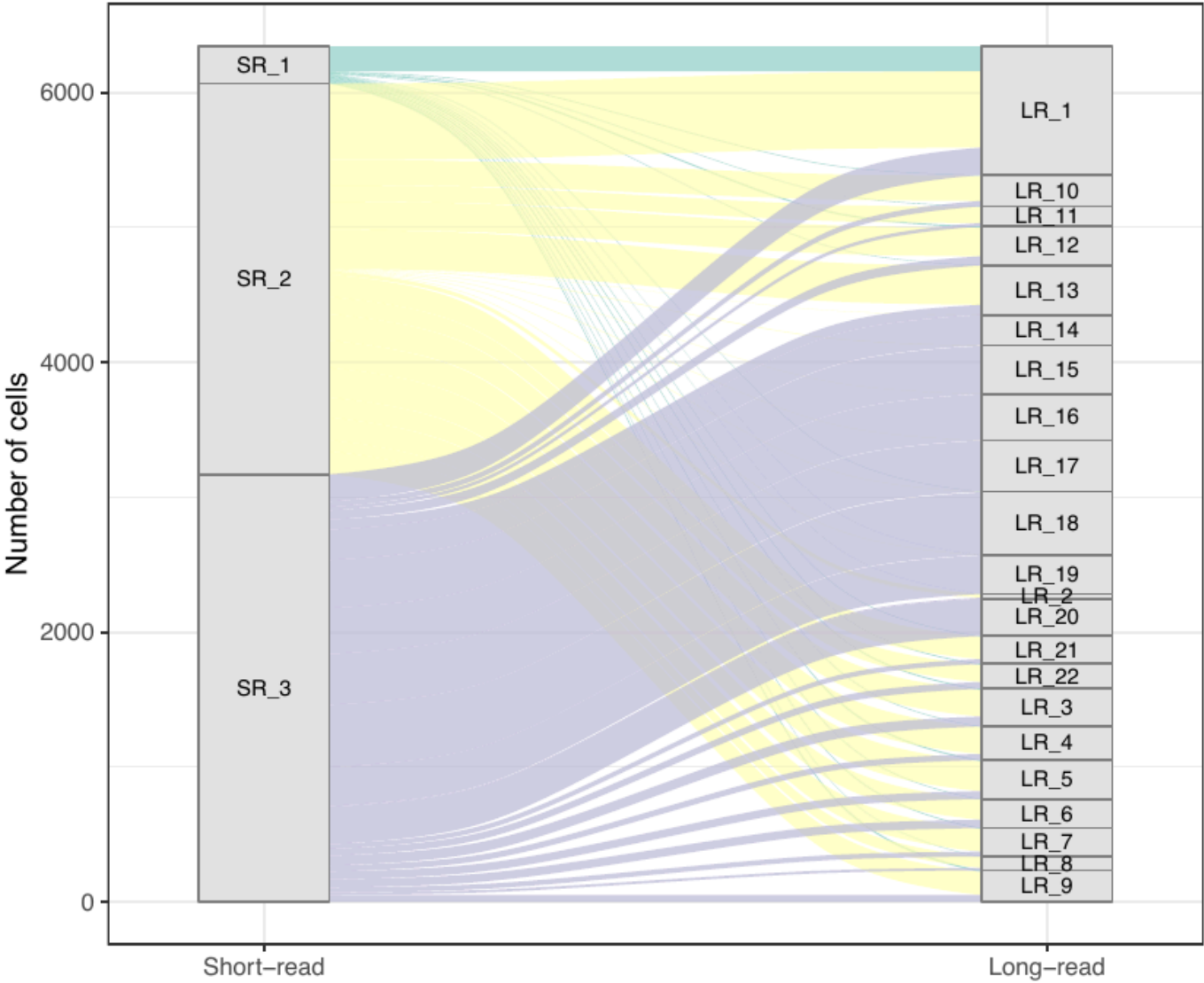
LK2_HBDN376-TP53_broad_lineage_composition_by_compartment.pdf

LK2_HBDN376-TP53 – Broad lineage composition by Numbat compartment

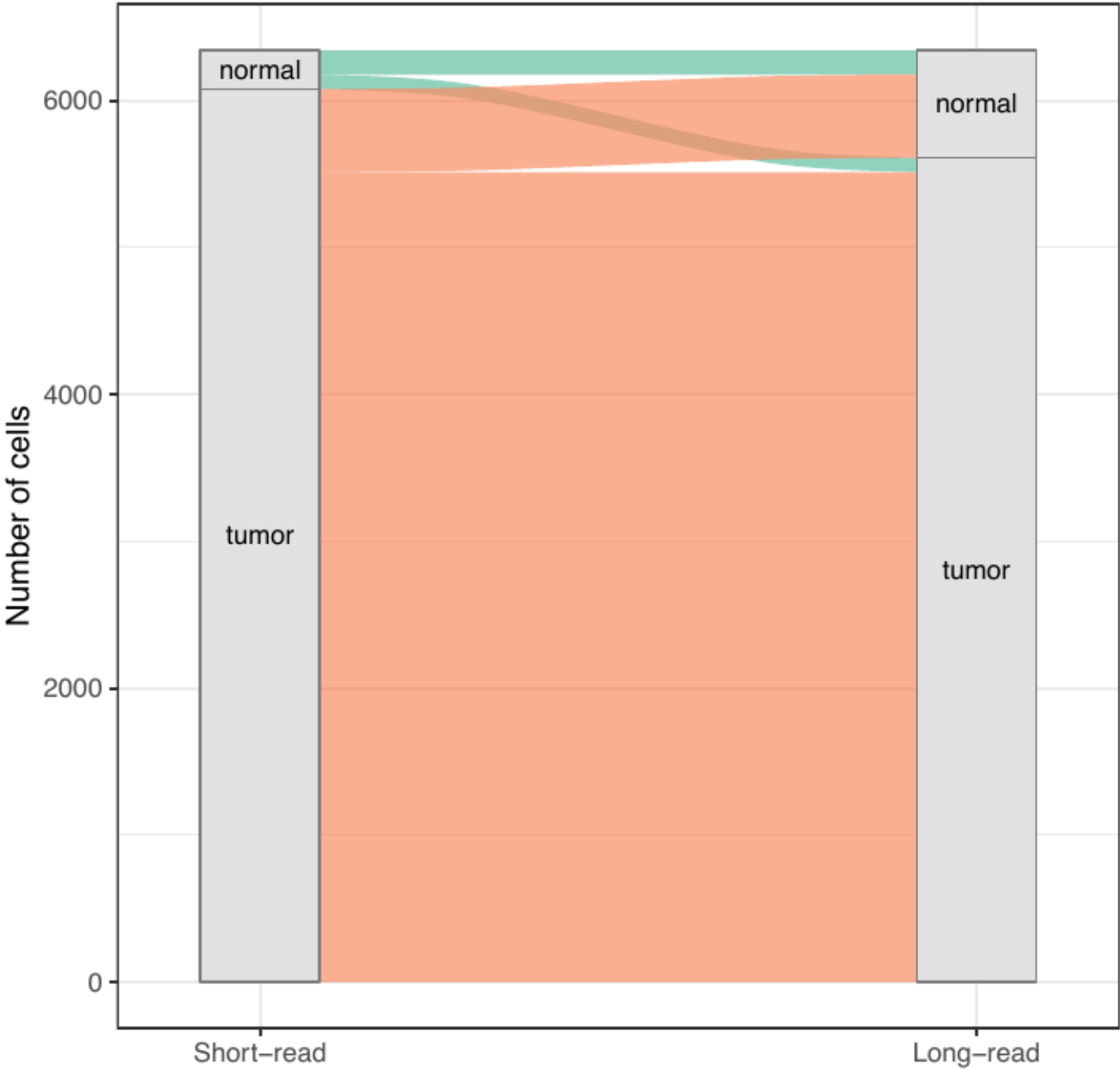


LK2_HBDN376-TP53_clone_alluvial.pdf

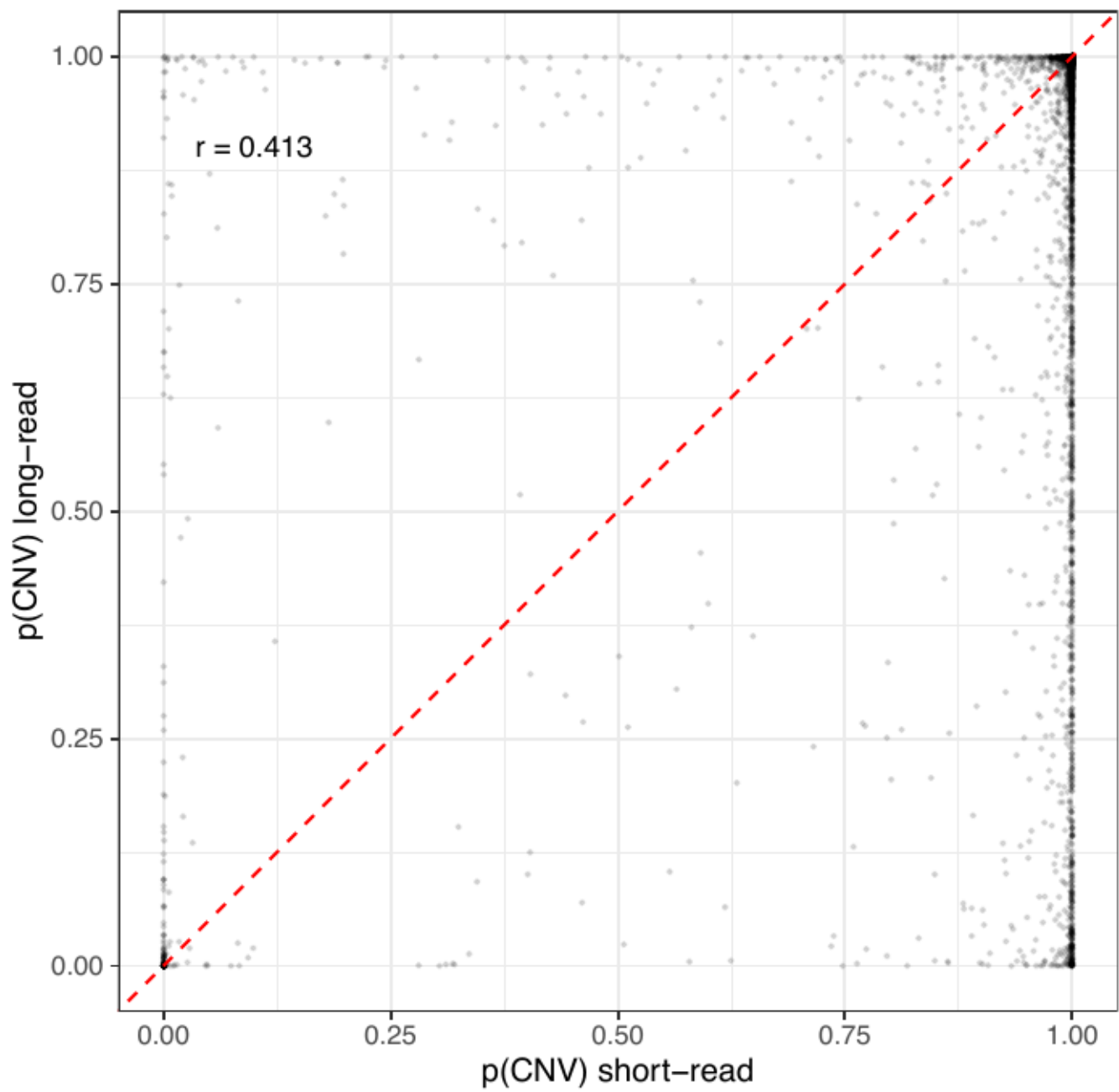
LK2_HBDN376-TP53 – Clone: short-read vs long-read



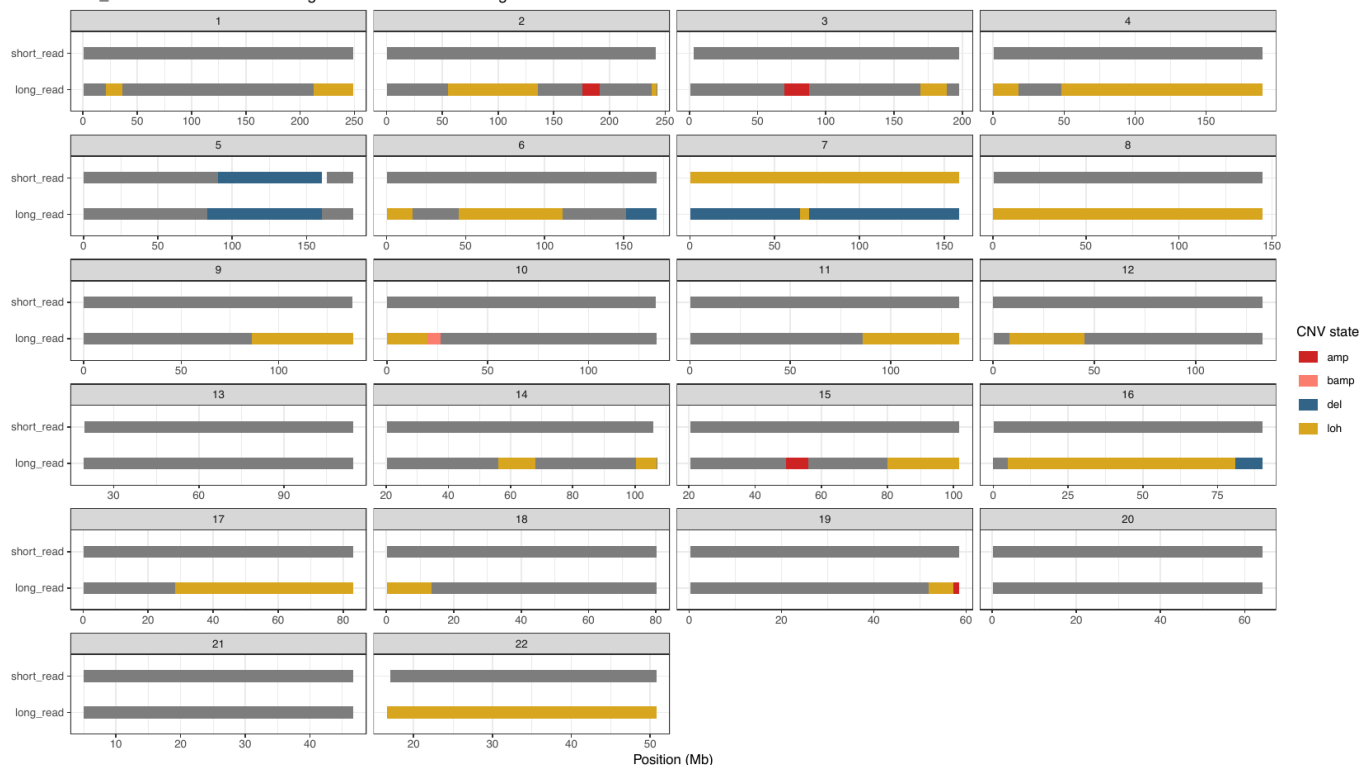
LK2_HBDN376-TP53 – Compartment: short-read vs long-re



LK2_HBDN376-TP53 – p(CNV) correlation

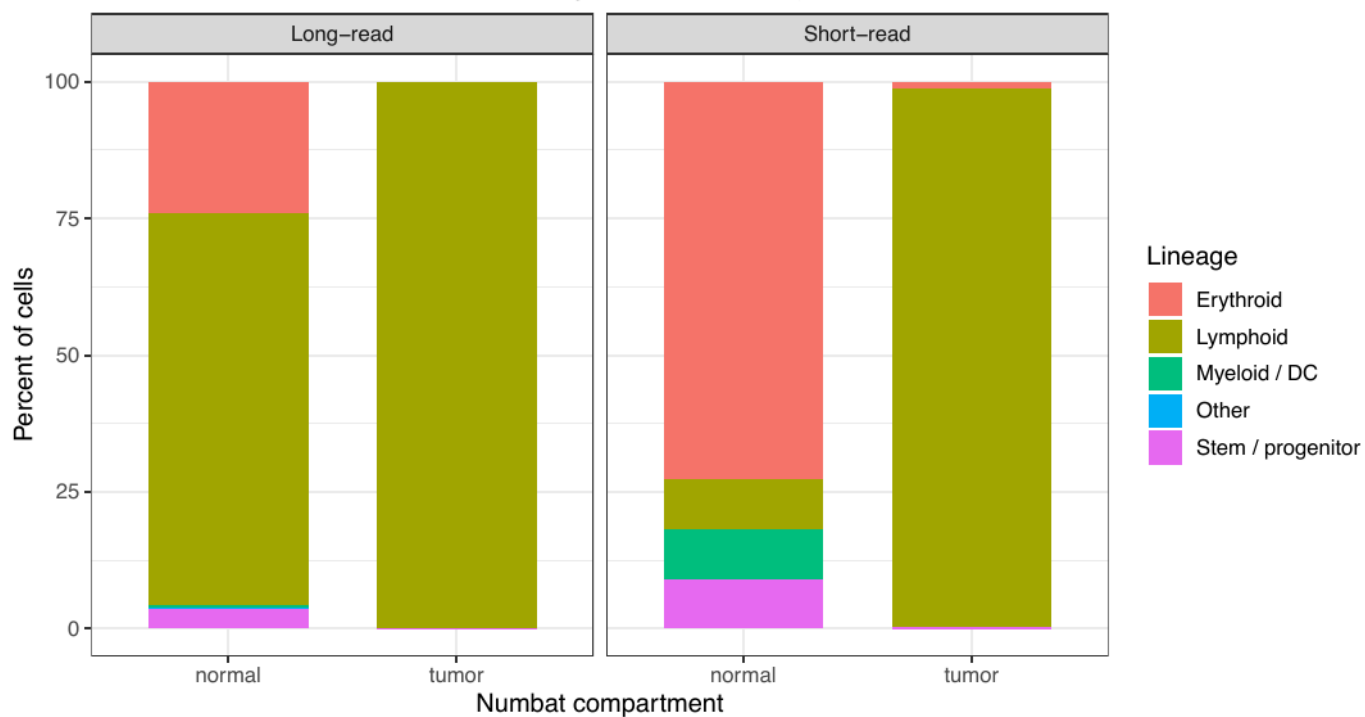


LK2_HBDN376-TP53 – CNV segments: short-read vs long-read



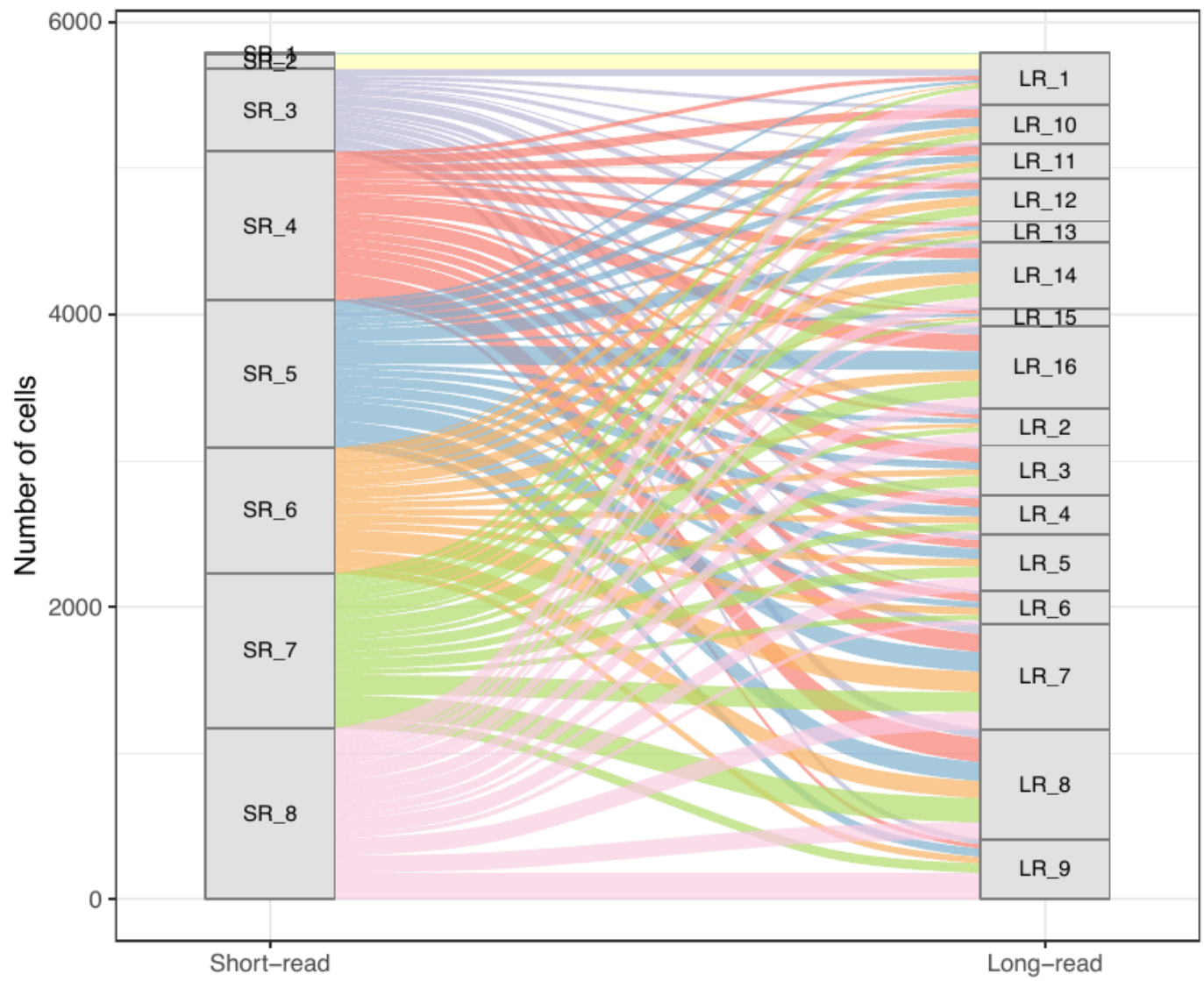
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LK2_HBDN498-TP53 – Broad lineage composition by Numbat compartment



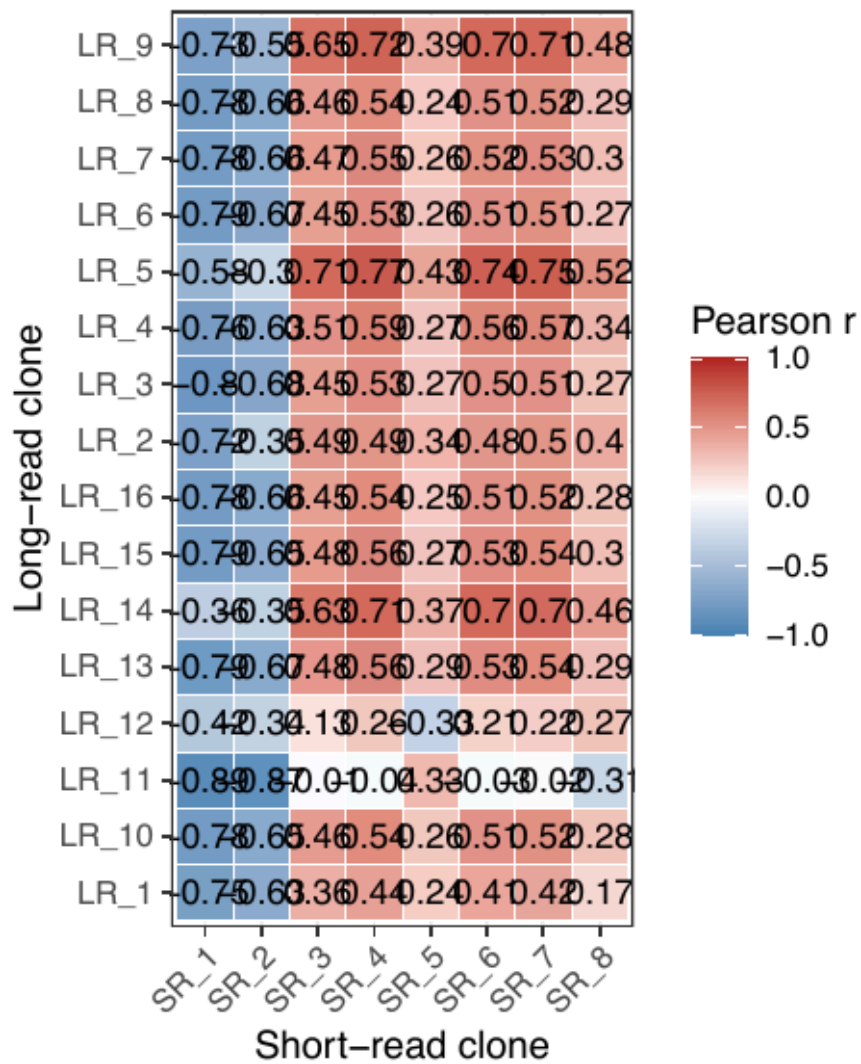
LK2_HBDN498-TP53_clone_alluvial.pdf

LK2_HBDN498-TP53 – Clone: short-read vs long-read

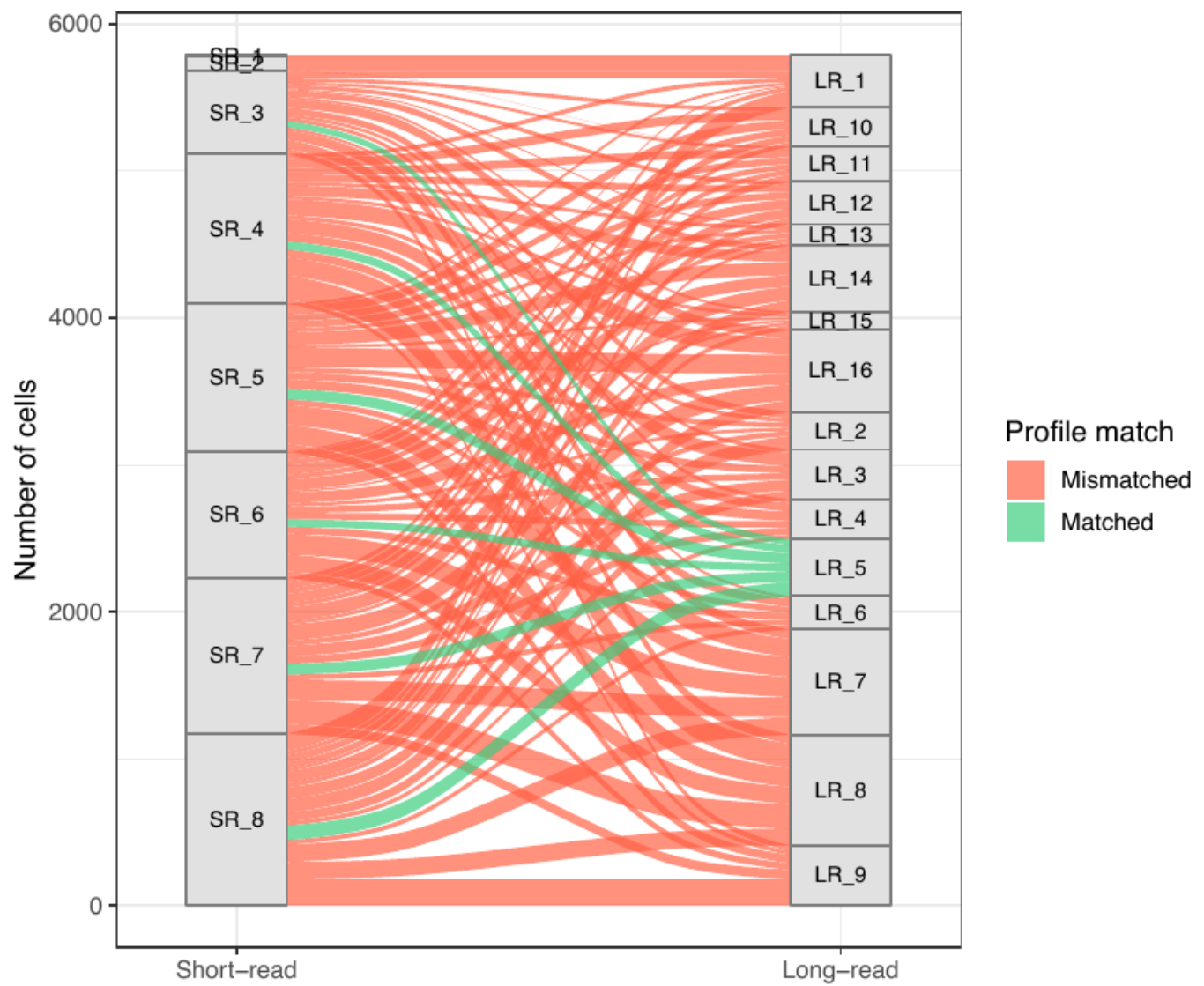


LK2_HBDN498-TP53_clone_correlation_heatmap.pdf

LK2_HBDN498-TP53 – Clone CNV profile corre

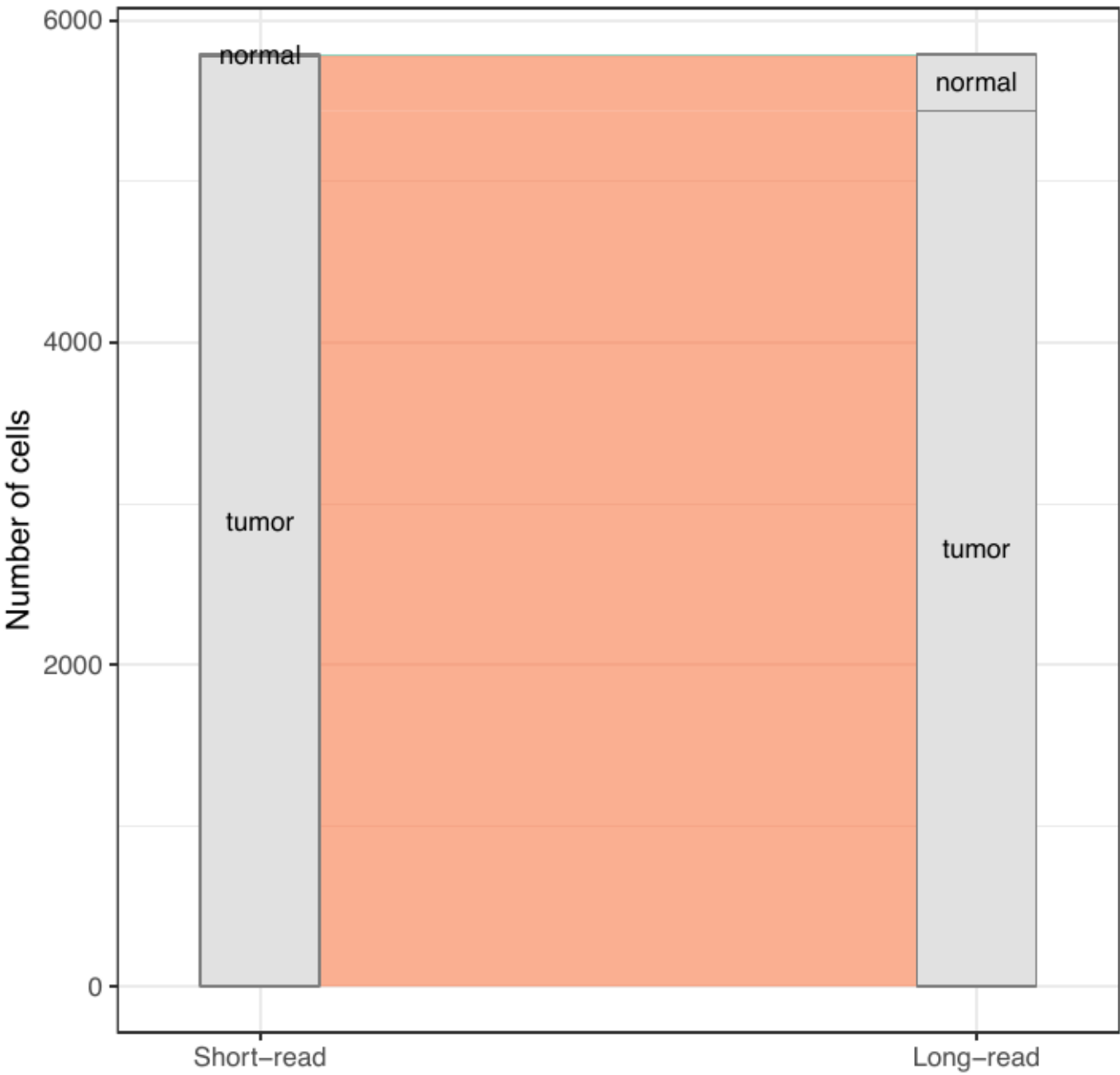


LK2_HBDN498-TP53 – Clone matching (by CNV profile)

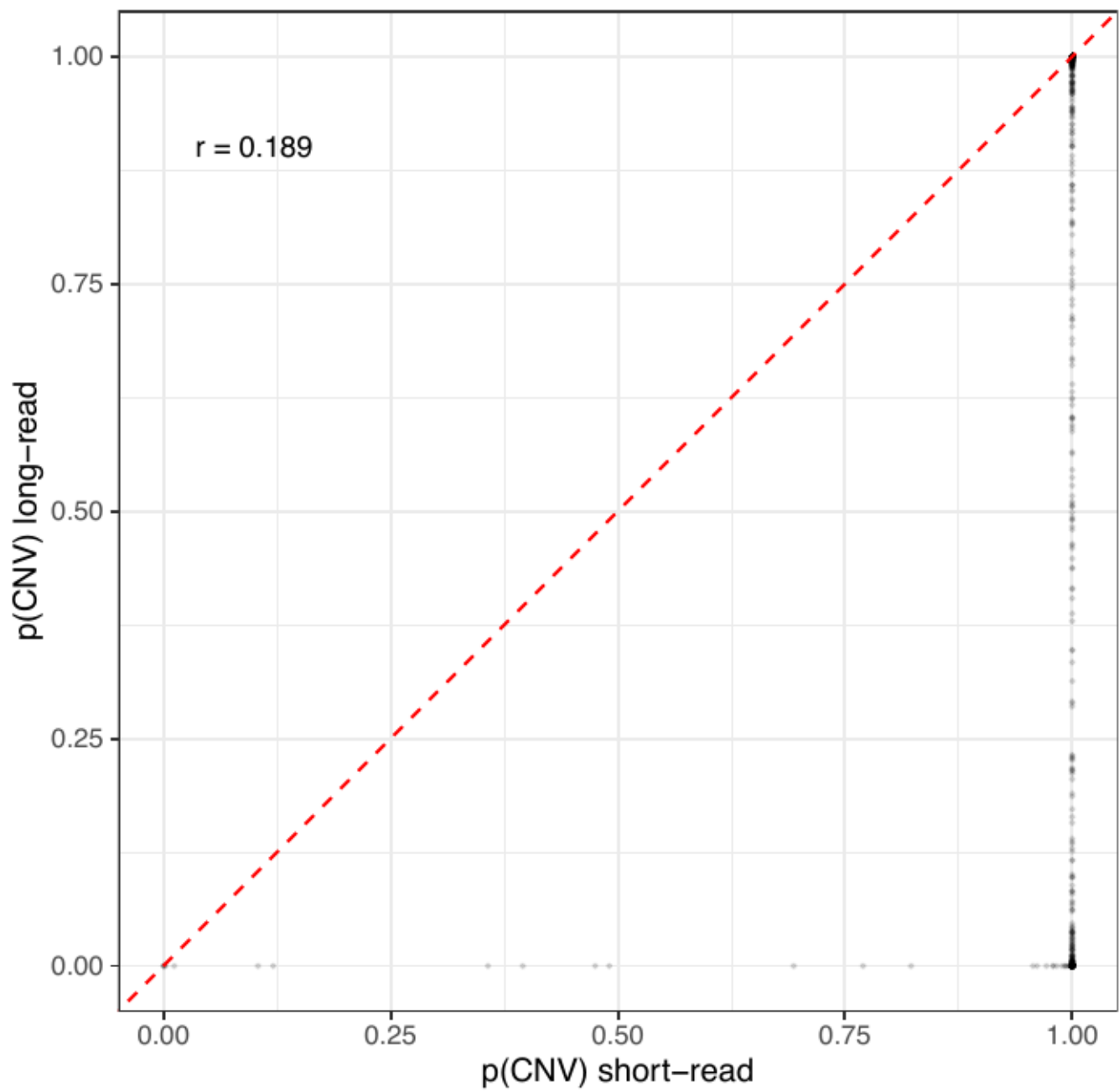


LK2_HBDN498-TP53_compartment_alluvial.pdf

LK2_HBDN498-TP53 – Compartment: short-read vs long-re



LK2_HBDN498-TP53 – p(CNV) correlation



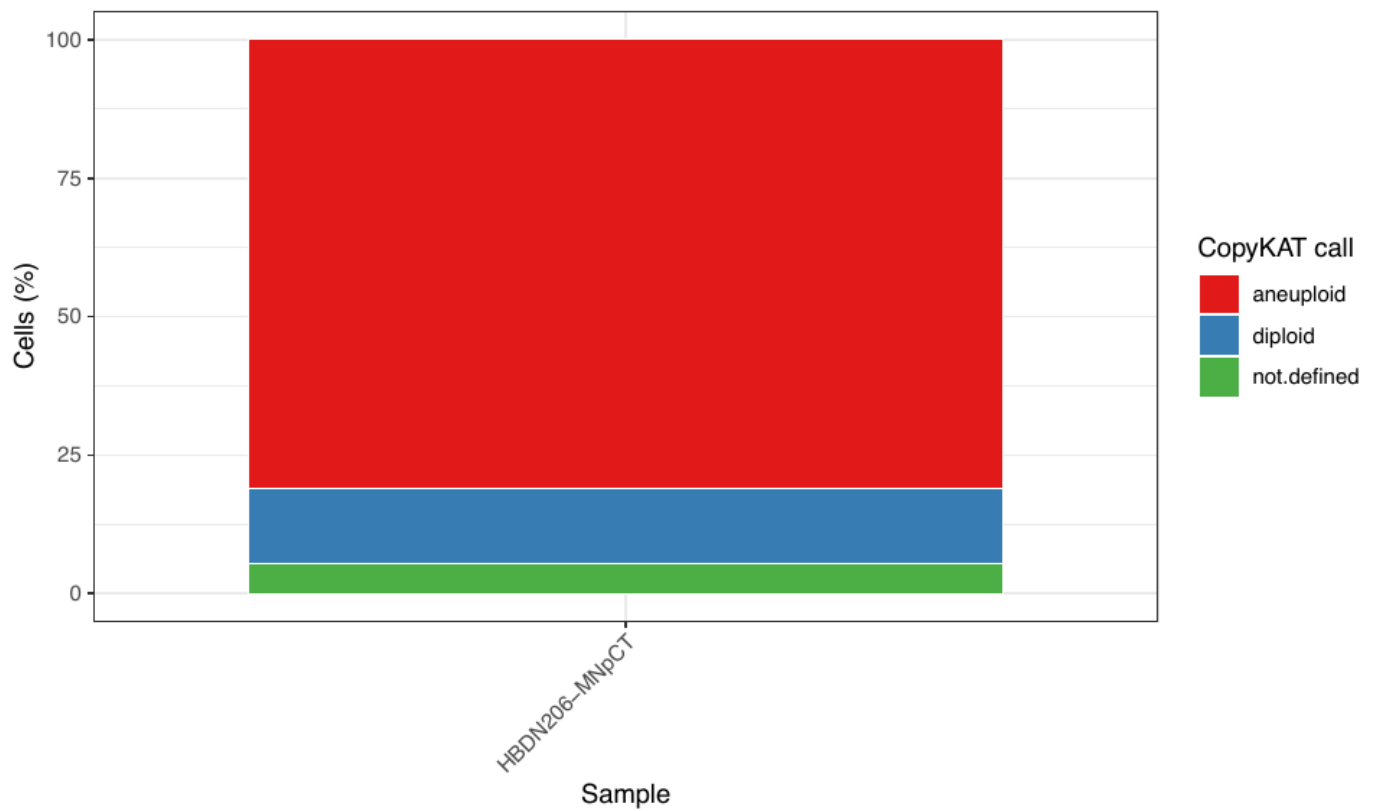
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LK2_HBDN498-TP53 – CNV segments: short-read vs long-read



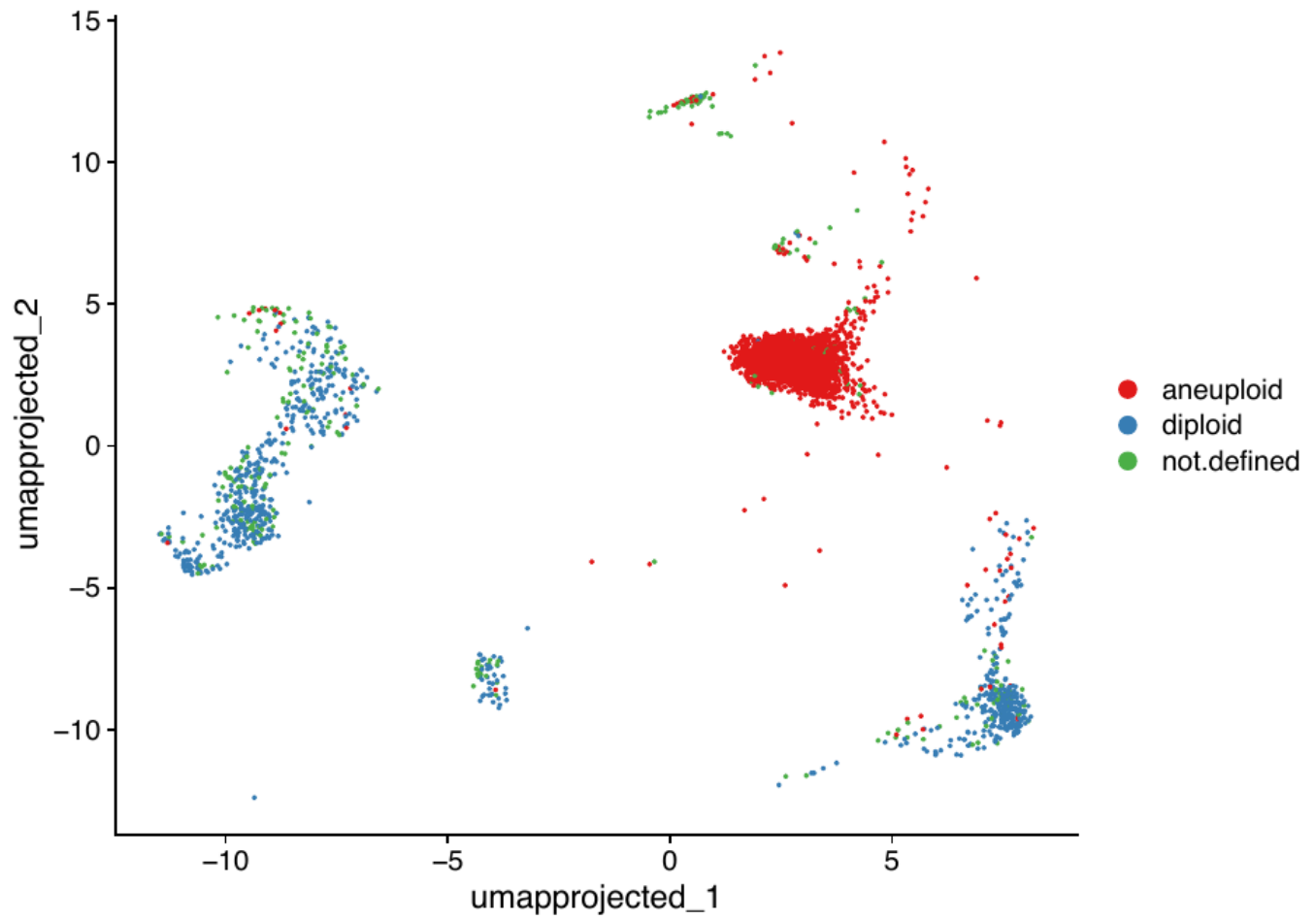
01_copykat_call_composition_by_sampleID.pdf

CopyKAT malignant/normal-like composition



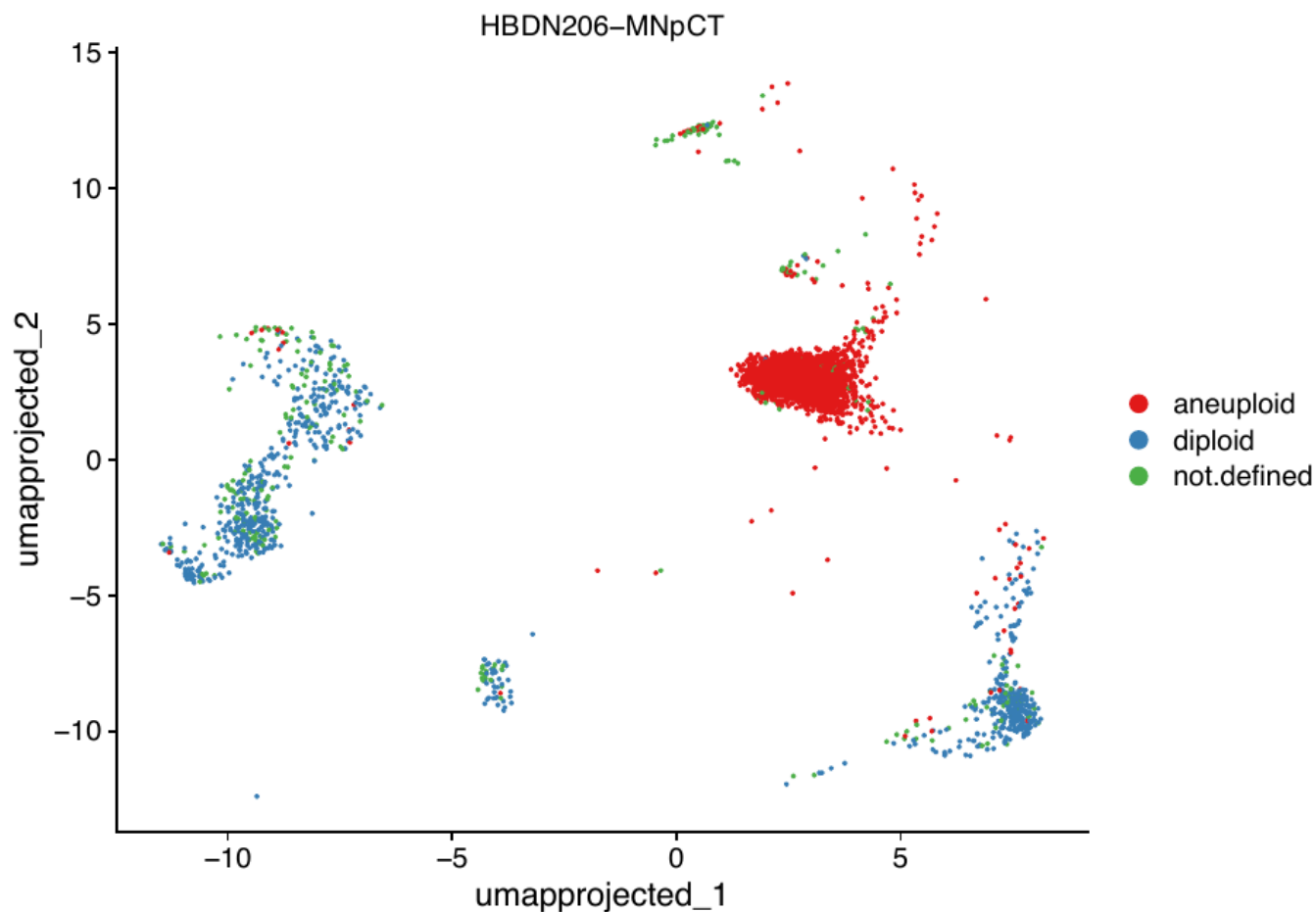
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CopyKAT calls on projected RNA UMAP

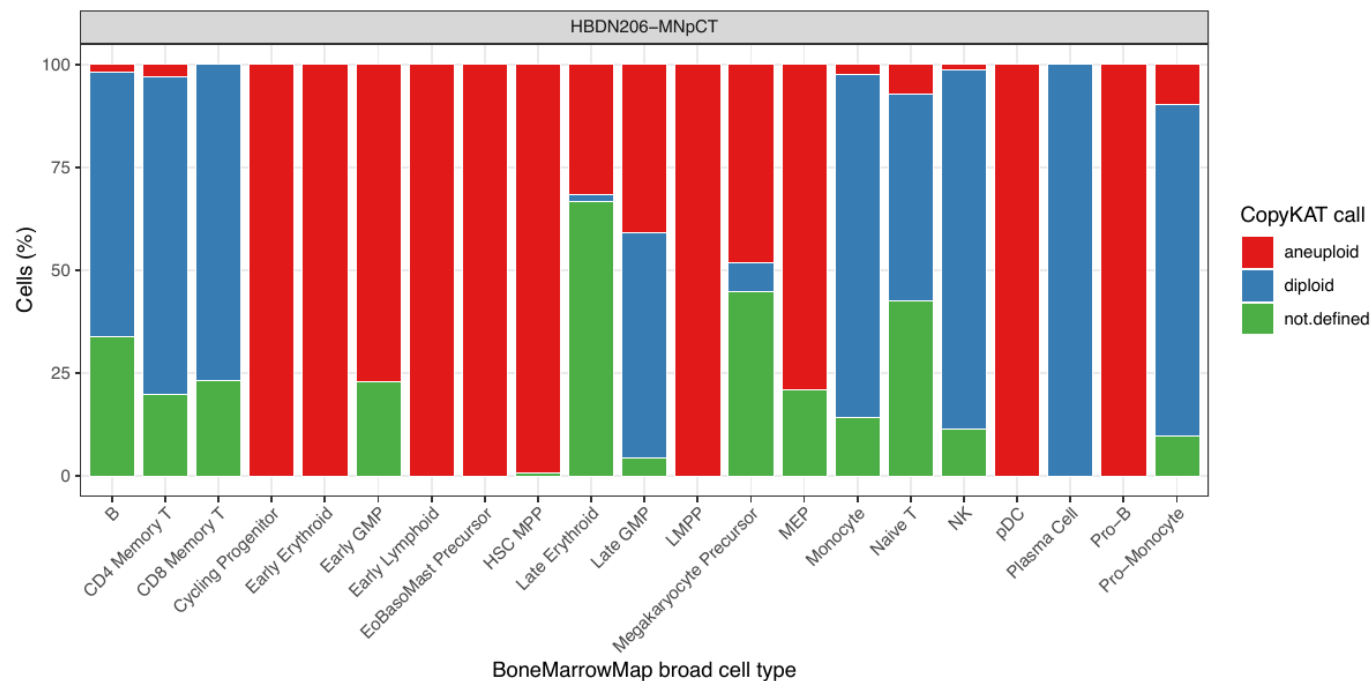


02a_copykat_calls_on_projected_RNA_UMAP.pdf

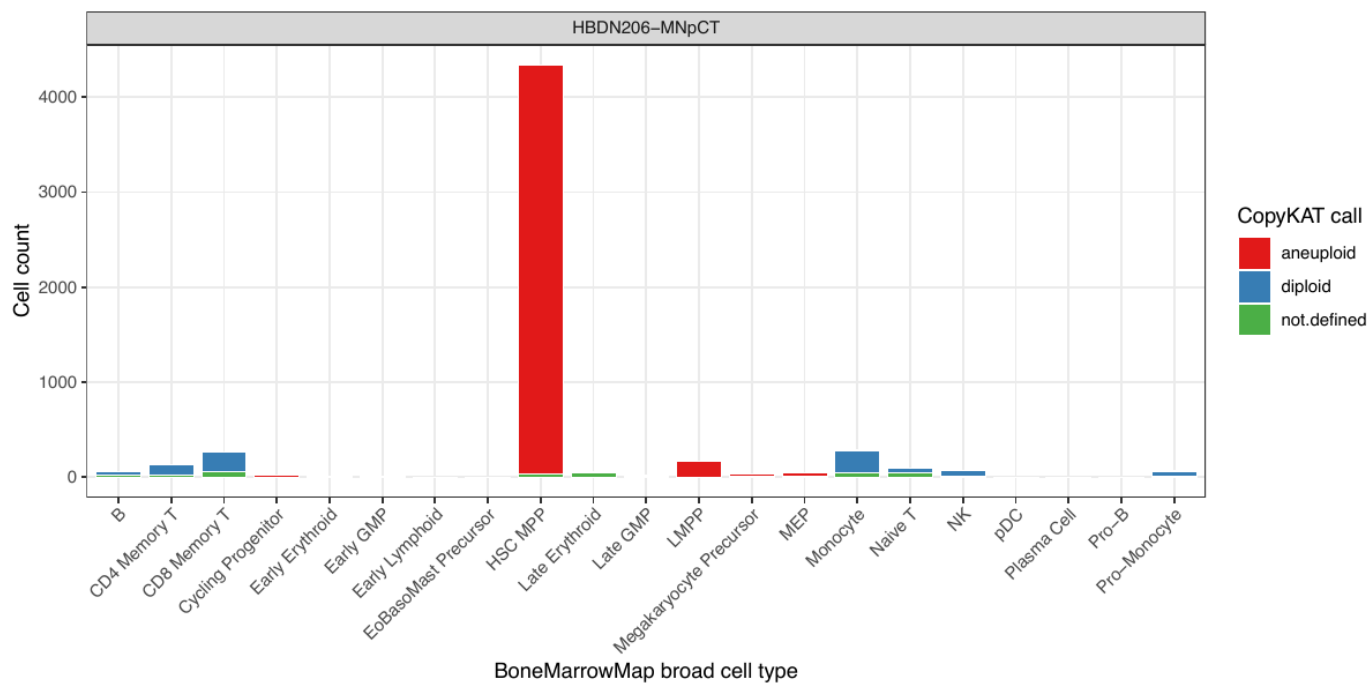
CopyKAT calls on projected RNA UMAP by sampleID



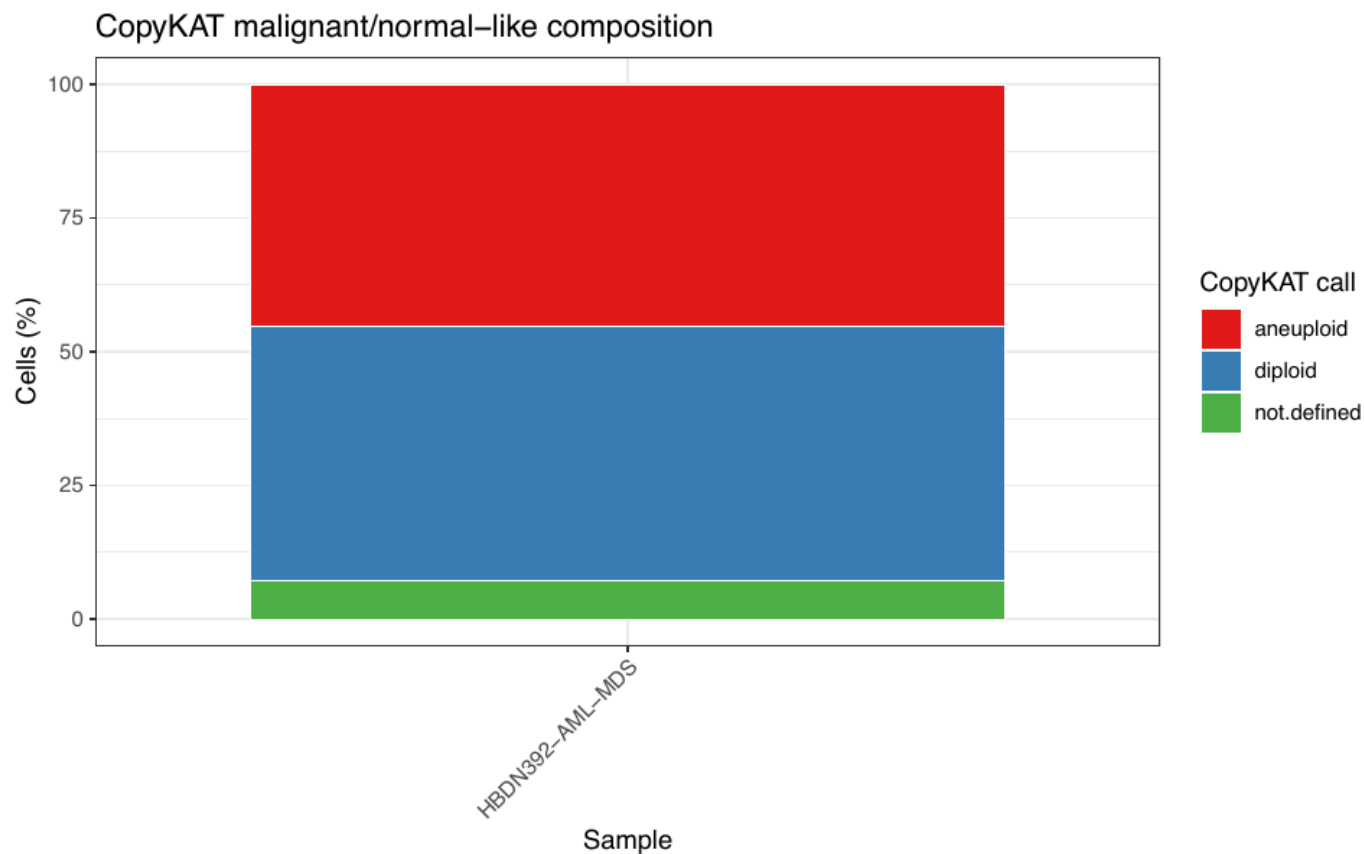
03_copykat_by_BoneMarrowMap_broad_celltype.pdf



03a_copykat_counts_by_BoneMarrowMap_broad_celltype.pdf

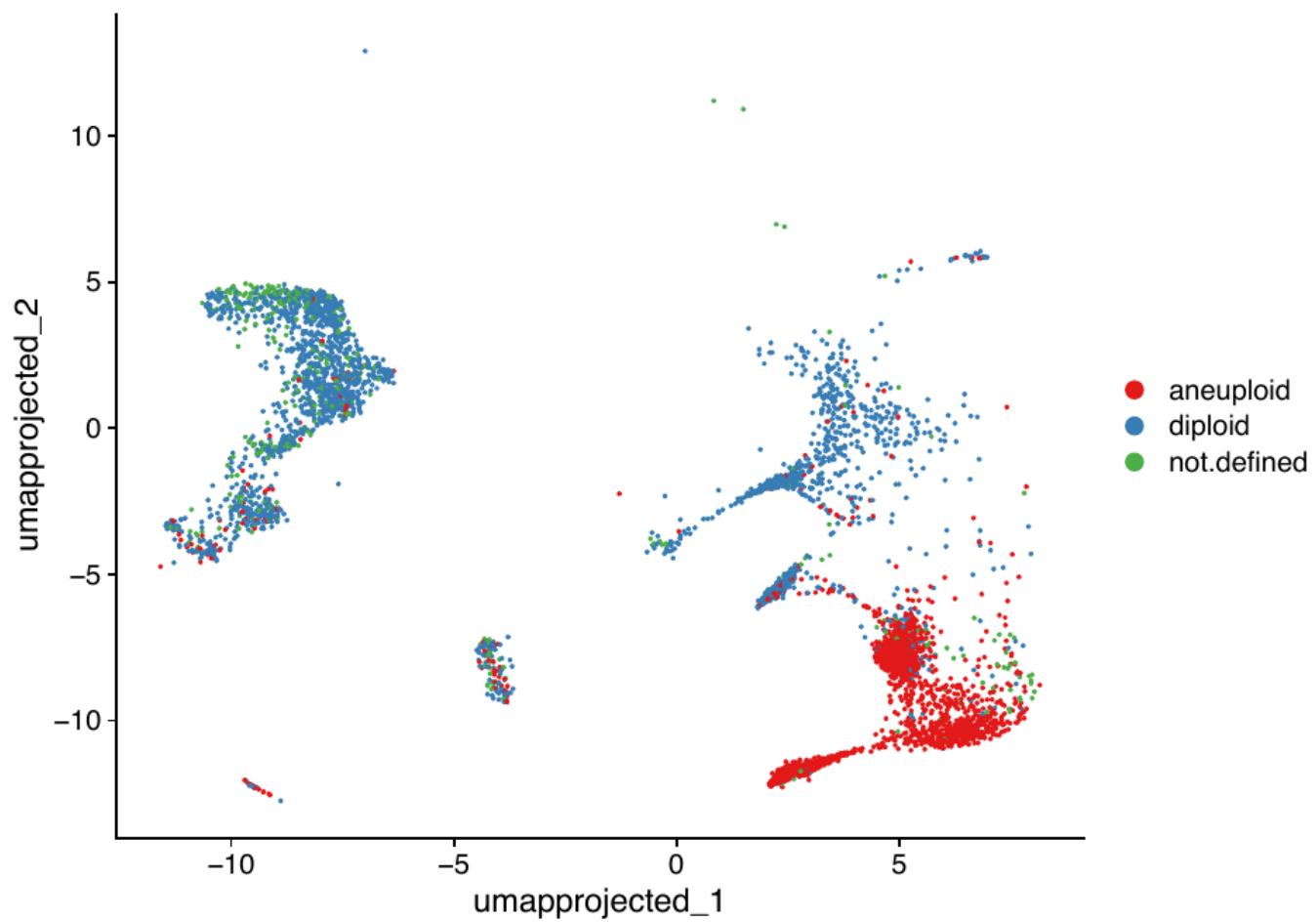


01_copykat_call_composition_by_sampleID.pdf



02_copykat_calls_on_projected_RNA_UMAP.pdf

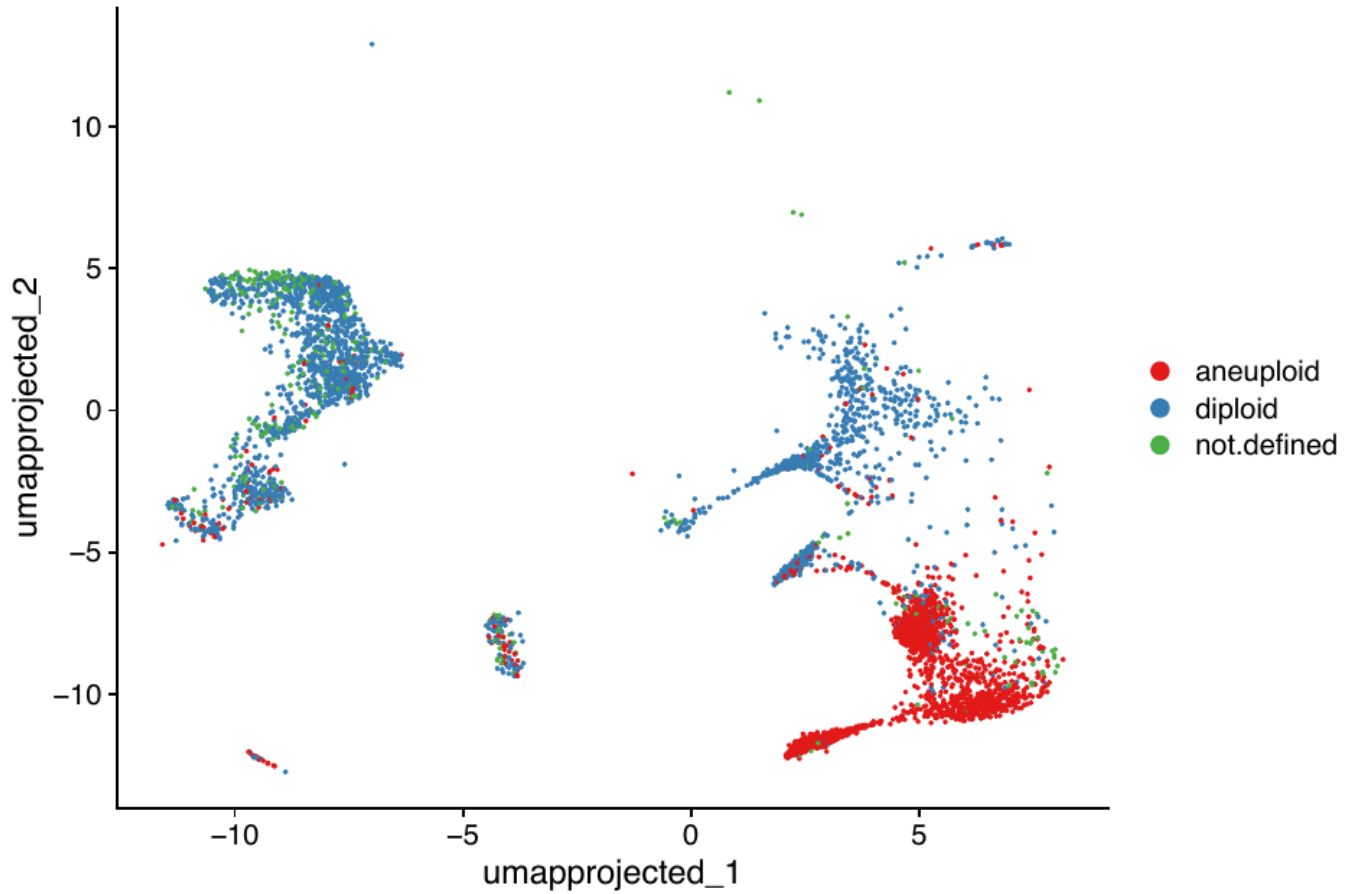
CopyKAT calls on projected RNA UMAP



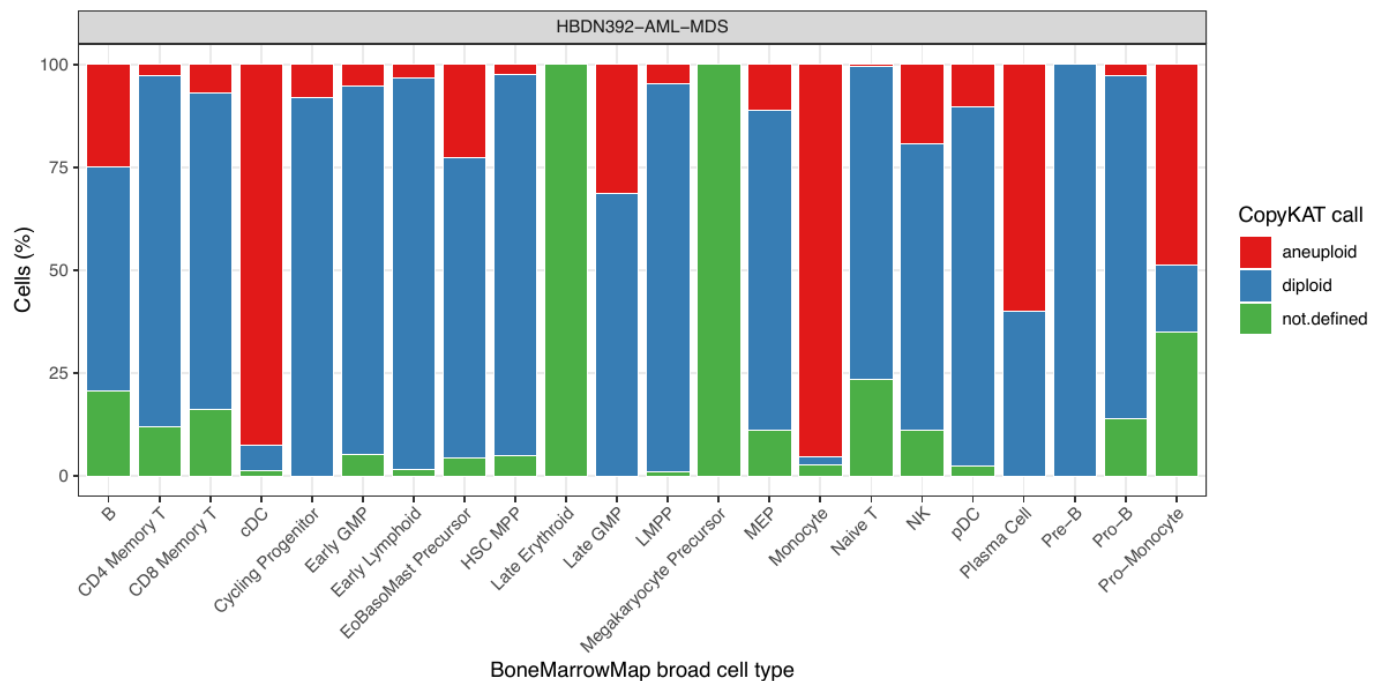
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CopyKAT calls on projected RNA UMAP by sampleID

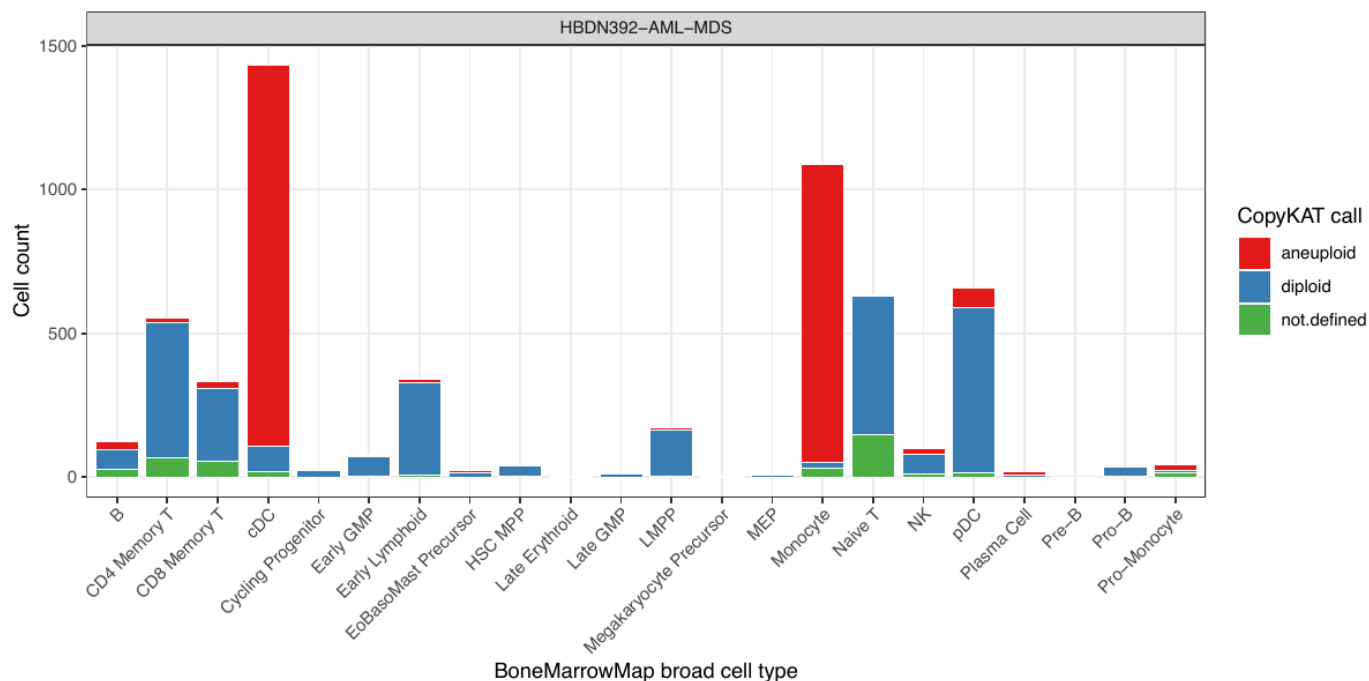
HBDN392-AML-MDS



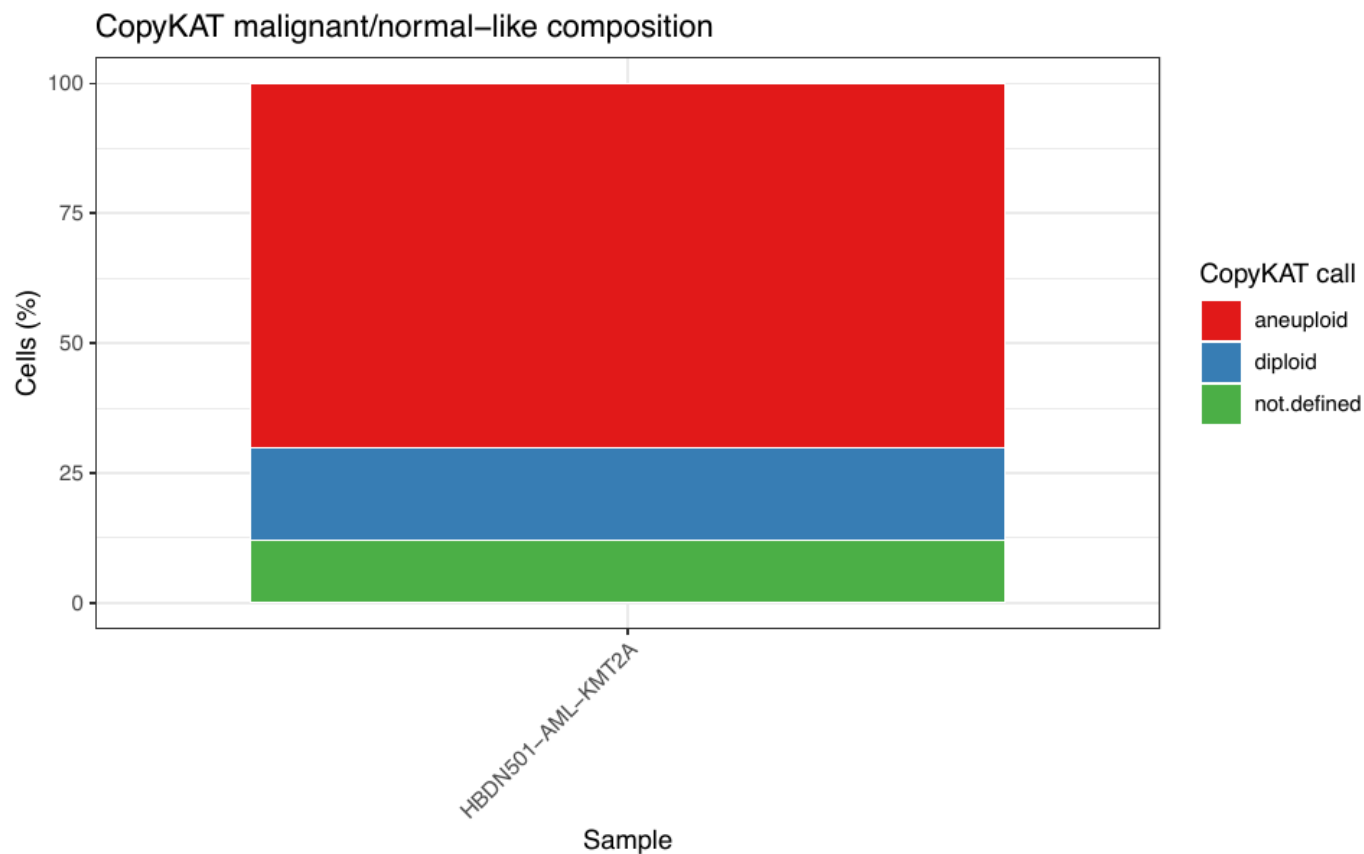
03_copykat_by_BoneMarrowMap_broad_celltype.pdf



03a_copykat_counts_by_BoneMarrowMap_broad_celltype.pdf

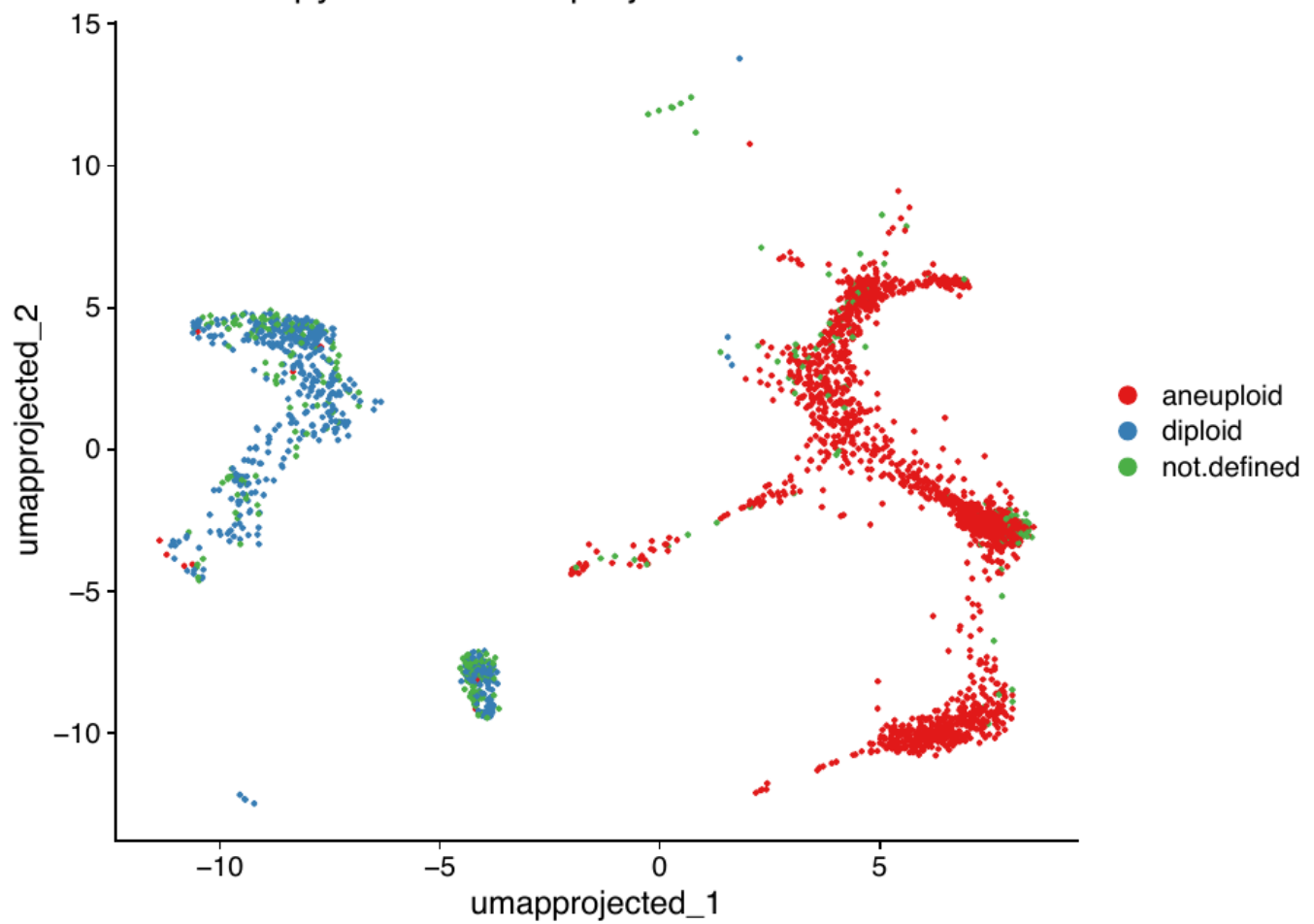


01_copykat_call_composition_by_sampleID.pdf



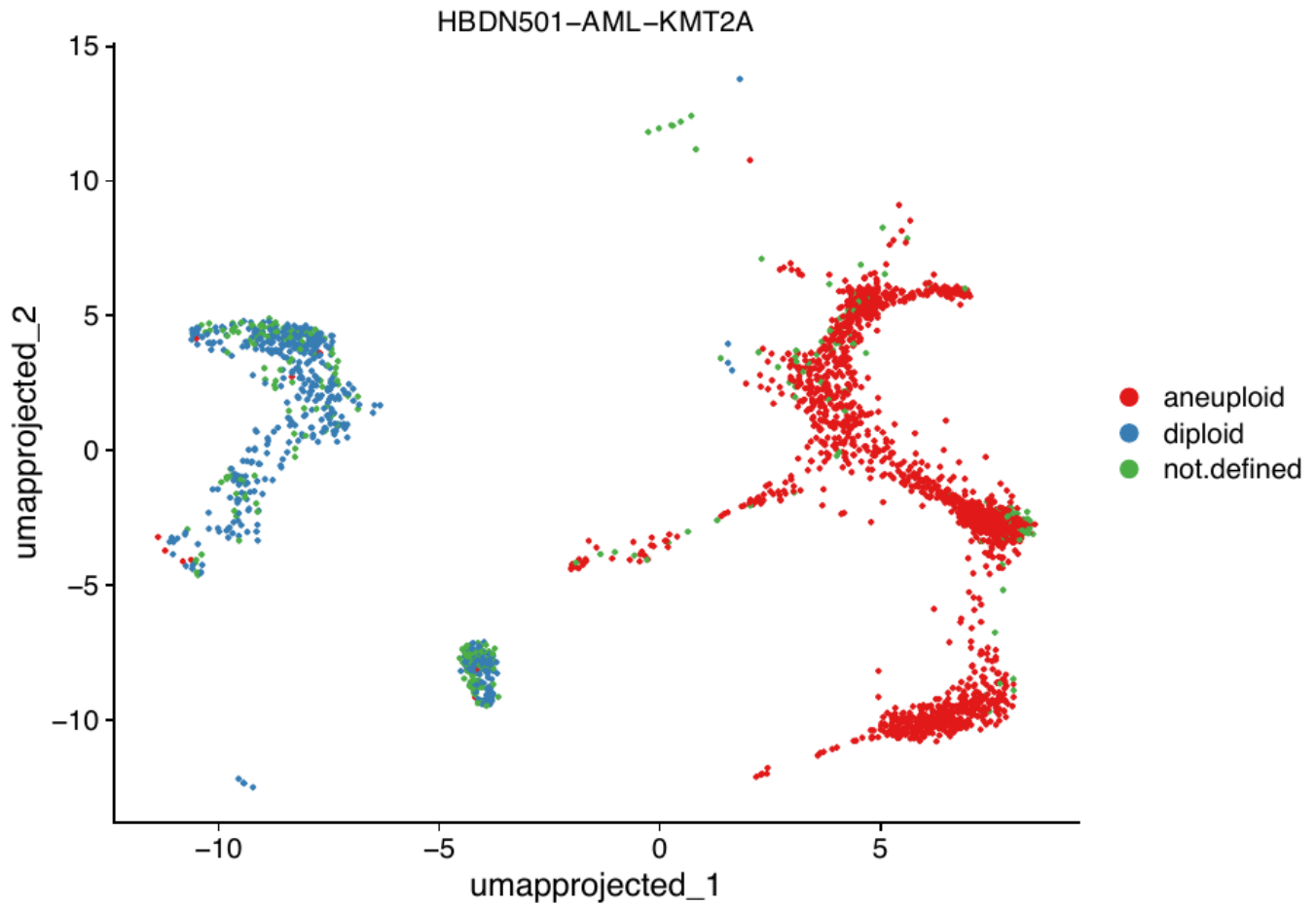
02_copykat_calls_on_projected_RNA_UMAP.pdf

CopyKAT calls on projected RNA UMAP

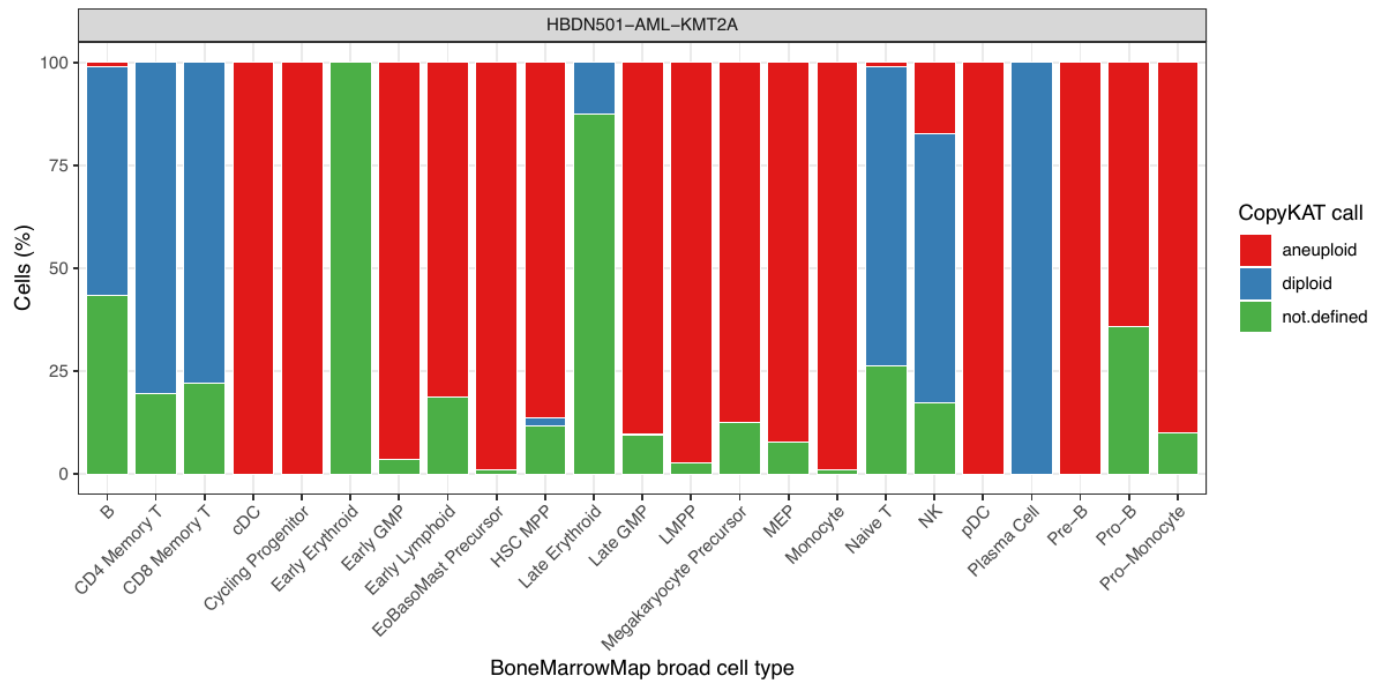


02a_copykat_calls_on_projected_RNA_UMAP.pdf

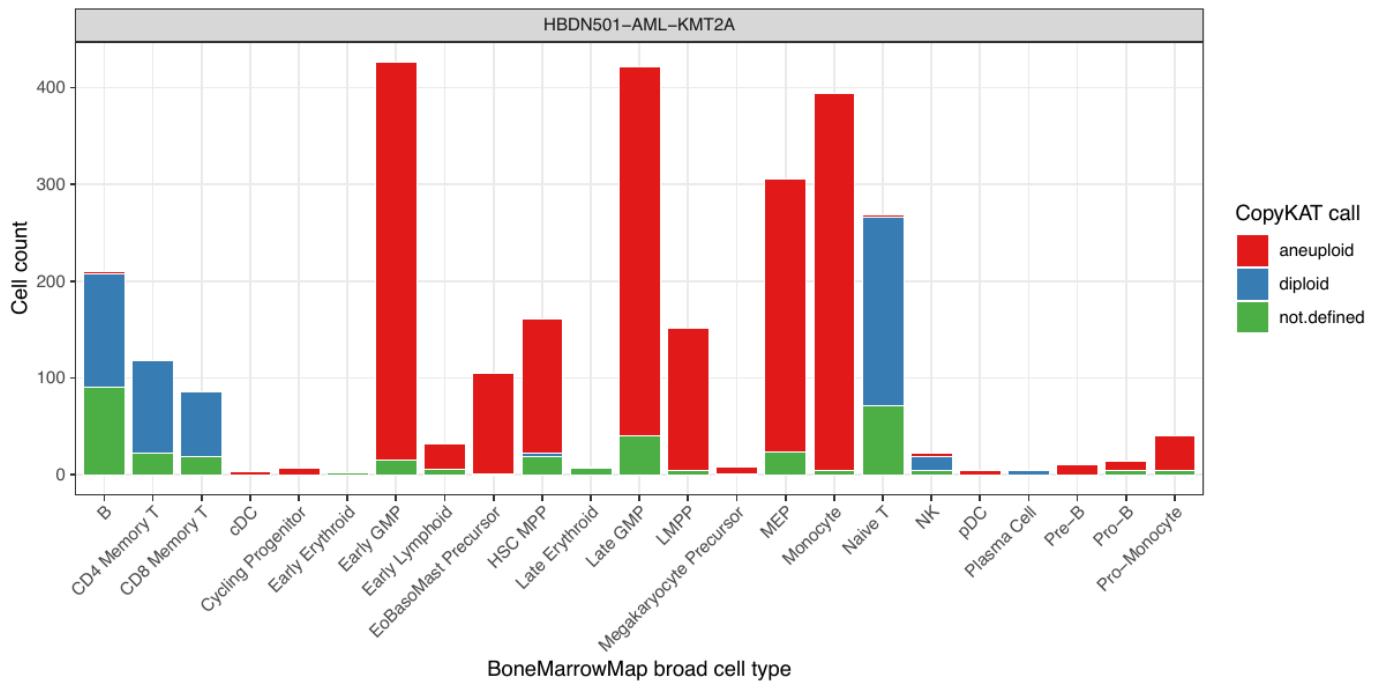
CopyKAT calls on projected RNA UMAP by sampleID



03_copykat_by_BoneMarrowMap_broad_celltype.pdf



03a_copykat_counts_by_BoneMarrowMap_broad_celltype.pdf



6. What was run

Step	Script	What
01	01.copykat_per_donor.R	CopyKAT per donor, lymphoid cells as the confident-normal anchor. MOLM13 assigned rather than called.
02	02.numbat_pileup_and_phase.sh	Numbat stage 1 on the cluster: pileup and phasing from the Cell Ranger BAM, in the container.
03	03.numbat_run.sh + 03a.numbat_run.R	Numbat stage 2: CNV and clone inference, one array task per donor.
04	04.numbat_to_seurat.R	Pulls the per-donor Numbat calls onto the annotated object and plots compartment and clone against the BoneMarrowMap annotation.
05	05.compare_numbat_copykat.R	Numbat against CopyKAT on one donor, with the clone inspection. Donor and doublet handling are arguments.
06	06.numbat_normal_donor_control.R	The specificity control: what Numbat reports on the healthy donor, against the three AML donors.
07	07.compare_shortread_longread_numbat.R	Short-read against long-read Numbat on the same cells.
08	08.cancer_calling_summary.R	Scores the callers against each other from the tables alone.
09	09.cancer_calling_report.Rmd	This report.

The scripts this was collated from, and what changed:

- 9. `copykat.R` ran CopyKAT over the whole GEM well at once. That is wrong here, not merely superseded - the well pools four donors and CopyKAT's baseline comes from the cells it is given, so one donor's aneuploidy could define another's normal. 9a. `copykat_perSample.R` is the per-donor version and is what produced the surviving outputs.
- the donor loop in 9a. `copykat_perSample.R` was `unique(temp$sample_name)[c(4)]`, a hardcoded index, so it did one donor per invocation with the index edited by hand between runs. It now loops over all of them.
- 9b. `numbat.sh` declared `SAMPLES` twice, LK1 first and LK2 second, so the second assignment silently overrode the first and the LK1 block was dead code that looked live. It also had `set -euo pipefail` commented out, so a failure inside the container did not fail the array task and a broken donor looked finished. Both fixed, and the run is now an argument.
- 10. `numbat_annotation.R` (LK2) and 13. `numbat_annotation.R` (LK1) are the same script at two stages. 13 hardcodes the donor directories with two commented out, and has its `readRDS` commented out so it only ran if the object was already in the session. 10 derives the directories, records which Numbat outputs each donor actually has, and handles incomplete runs. Step 04 is 10, parameterised by run.
- 12c. `compare_numbat_copykat.R`, 12c. `compare_numbat_copykat_501.R` and 12d. `compare_numbat_copykat_noDoublets.R` are one script run three times with the donor edited in place. The entire difference between 12c and 12d is one `scDbFinder` subset call and a different output directory name, which is now a flag.
- 12d. `investigate_clones.R` is **not** a separate analysis - 326 of its 333 non-comment lines are verbatim inside 12d. `noDoublets`. It is the earlier standalone version, already folded in, so step 05 covers it.
- 20a. `compare_all_clone_methods.R` is deliberately out of scope: it also brings in the mitochondrial clone callers and belongs with that report.

7. Data locations

Tools

What	Where
Annotated input, pilot	results/seurat_annotated/260423_VH01624_453_222HWMYNNX/LK1_GEX_CITE_DSB_BoneMarrowMap_projected.rds
Annotated input, LK2	results/seurat_annotated/260528_VH01624_464_222K7VKNX/demux_singletons_annotated_seurat.rds
Cell Ranger BAM (Numbat input)	results/cellranger_withbam//GEX/outs/possorted_genome_bam.bam
Phased SNP panel	annotation/genome1K.phase3.SNP_AF5e2.chr1toX.hg38.sorted.vcf.gz
Numbat container	containers/numbat-rbase_latest.sif
CopyKAT output	results/seurat_annotated/260423_VH01624_453_222HWMYNNX/copykat_annotated/
Numbat output, pilot	results/seurat_annotated/260423_VH01624_453_222HWMYNNX/numbat/LK1_/numbat_final
Numbat output, LK2	results/seurat_annotated/260528_VH01624_464_222K7VKNX/numbat/LK2_/numbat_final
Numbat output, pipeline	results_nf/260528_VH01624_464_222K7VKNX/09c_numbat_run
Long-read comparison	results/longread_numbat/comparison_figures, comparison_tables
Benchmark tables and figures (step 08)	results/benchmarking/cancer_calling

Tool	Version	Use
CopyKAT	see sessionInfo	expression-only CNV calling, per donor
Numbat	container numbat-rbase_latest.sif - unpinned latest tag, version not recorded	allele-aware CNV and clone inference
Eagle	v2.4.1, inside the Numbat container	phasing
Seurat	5.4.0	object handling and plotting

8. Caveats

- steps 01-05 and 07 are the collated, cleaned versions of the scripts that produced the figures shown here; they were not re-executed for this report. Steps 06 and 08 were run fresh, which is what `Analysis finished` reflects
- the Numbat container is pinned to the `latest` tag, so the exact version that produced these results is not recorded and cannot be reconstructed from the image name. Any re-run may not reproduce them
- **Numbat completed for only two of the four pilot donors.** HBDN392-AML-MDS and MOLM13 are entirely `not_called`, so the caller comparison rests on two donors
- the CopyKAT/Numbat agreement is not an independent comparison in the lymphoid compartment, because those cells are CopyKAT's normal anchor. It is independent in the myeloid and progenitor compartments
- **HBDN206-MNpCT is called 81% aneuploid by CopyKAT and 74% tumour by Numbat**, and its name does not carry an AML designation the way the other pilot donors' do. Either it is a malignant sample that is simply not named as one, or both callers are failing on it in the same way. Nothing here settles which, and it should be checked against the sample sheet before these fractions are quoted
- the long-read comparison covers three LK2 donors only, and the `donorN` directories in the long-read output are positional - the mapping to sample names is an assumption, now asserted in step 07 rather than trusted silently
- step 06 reads the pipeline Numbat output under `results_nf/`, not the standalone output the other steps use. They are separate runs and their numbers are not interchangeable